哈爾濱工業大學(深圳)
HARBIN INSTITUTE OF TECHNOLOGY, SHENZHEN

Chapter 2 Planar Force Systems

Instructor: Tongtong GUO
College of Science



Chapter 2. Planar force system

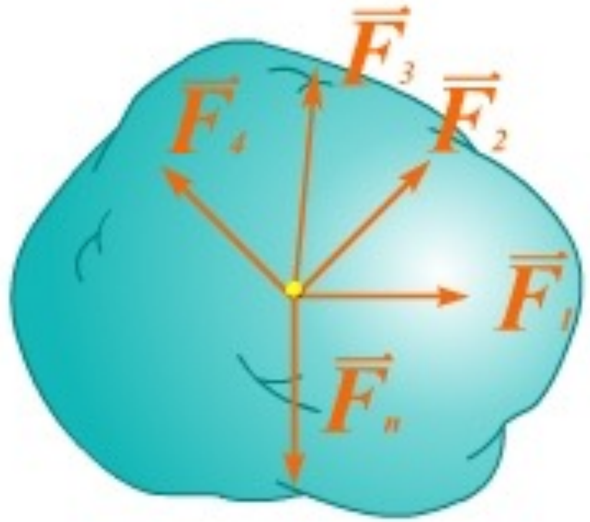
- 2.1 Coplanar concurrent force system
- 2.2 Moment of a planar force about a point
- 2.3 The equilibrium conditions and equilibrium equations for a planar arbitrary force system
- 2.4 Internal Force Calculation of Simple Plane Trusses

Chapter 2. Planar force system

- **2.1 Coplanar concurrent force system**
- **2.2 Moment of a planar force about a point**
- **2.3 The equilibrium conditions and equilibrium equations for a planar arbitrary force system**
- **2.4 Internal Force Calculation of Simple Plane Trusses**

• 2.1 Coplanar concurrent force system

I、The geometric method for the composition of a coplanar concurrent force system—the force polygon rule.

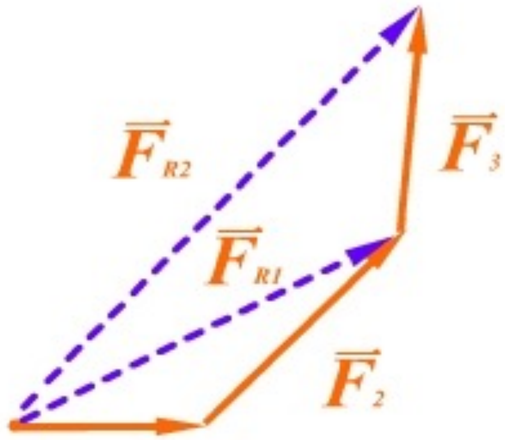


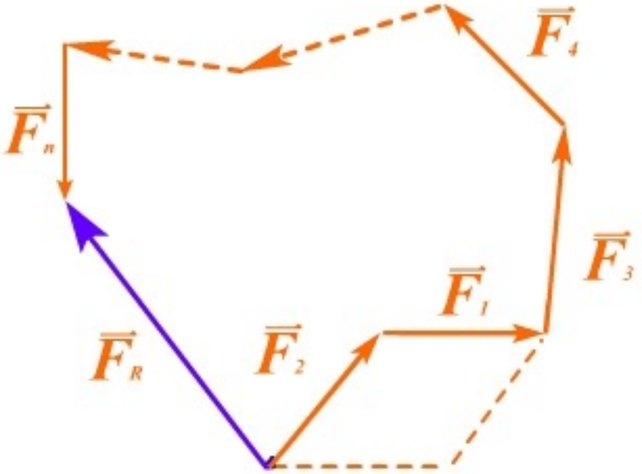
• 2.1 Coplanar concurrent force system

$$\boldsymbol{F}_{R1} = \boldsymbol{F}_1 + \boldsymbol{F}_2$$

$$\boldsymbol{F}_{R2} = \boldsymbol{F}_{R1} + \boldsymbol{F}_3$$

$$\boldsymbol{F}_R = \boldsymbol{F}_1 + \boldsymbol{F}_2 + \ldots + \boldsymbol{F}_n = \sum_{i=1}^{n} F_i$$





$$F_R = \sum_{i=1}^{n} F_i$$

Force Polygon

Force Polygon Rule

• 2.1 Coplanar concurrent force system

II、Geometric conditions for equilibrium of coplanar concurrent force systems

conditions for equilibrium

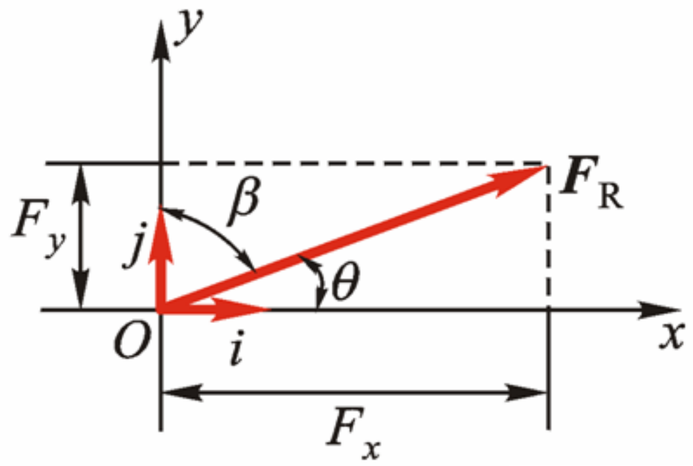
$$\sum_{i=1}^{n} F_i = 0$$

The necessary and sufficient conditions for the equilibrium of coplanar cocurrent force systems are:

he force polygon of the system closes by itself.

• 2.1 Coplanar concurrent force system

III、**The analytical method for the composition of a coplanar concurrent force system.**



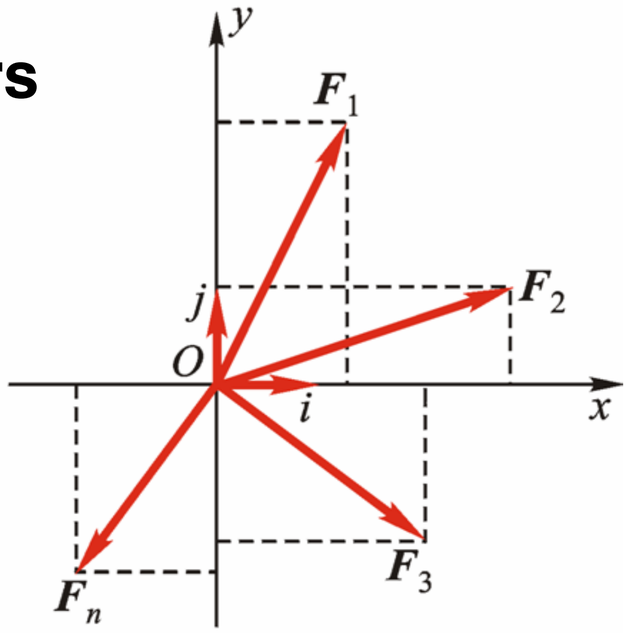
The joint force F_R is projected on the x-axis and y-axis,

$$F_{Rx} = F_R \cos\theta$$

$$F_{Ry} = F_R \cos\beta$$

The resultant force is equal to the sum of all force vectors

$$F_R = \sum_{i=1}^{n} F_i$$



The projection theorem of resultant force is derived from the projection theorem of combined vectors

$$F_{Rx} = \sum F_{ix} \qquad F_{Ry} = \sum F_{iy}$$

The magnitude of the resultant force is: $F_{\mathrm{R}} = \sqrt{F_{\mathrm{R}x}^{\ 2} + F_{\mathrm{R}y}^{\ 2}}$

The direction is $\cos(F_R, i) = \dfrac{\Sigma F_{ix}}{F_R}$ $\cos(F_R, j) = \dfrac{\Sigma F_{iy}}{F_R}$

The point of action is the convergence point of force

IV、The equilibrium equation of planar concurrent force system

Equilibrium condition $F_R = 0$

Equilibrium equation $\Sigma F_x = 0$ $\Sigma F_y = 0$

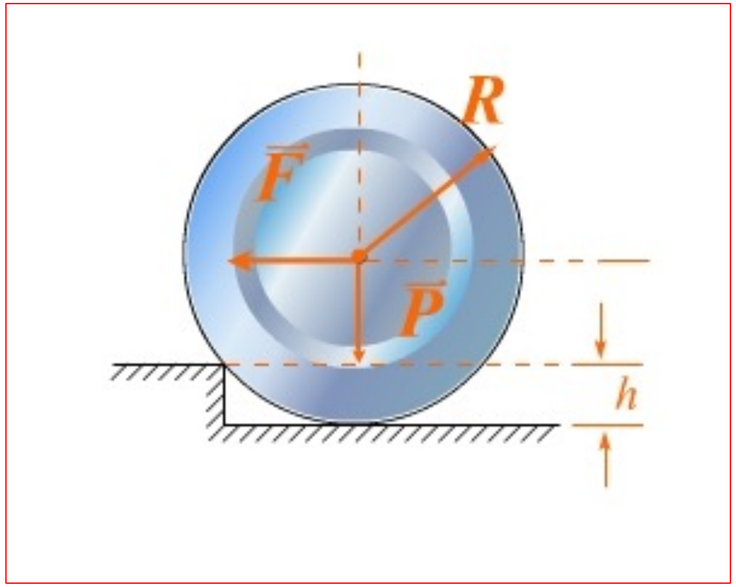
Example 2-1

Known: $P = 20\,\mathrm{kN}, R = 0.6\,\mathrm{m}, h = 0.08\,\mathrm{m}$ **solve（geometric method）：**

1. When the horizontal pulling force is 5 kN, what are the pressures exerted by the roller on the ground and the obstacle, respectively?

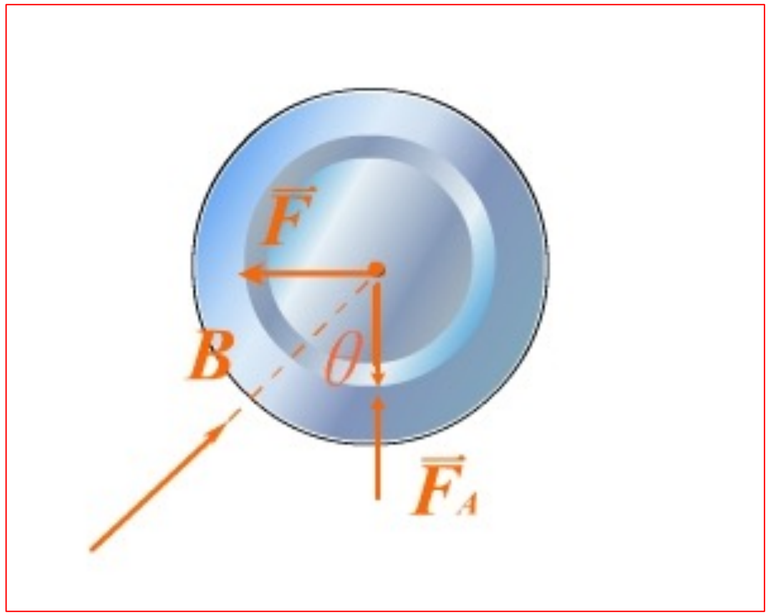
2. What is the minimum horizontal force F required to pull the roller over the obstacle?

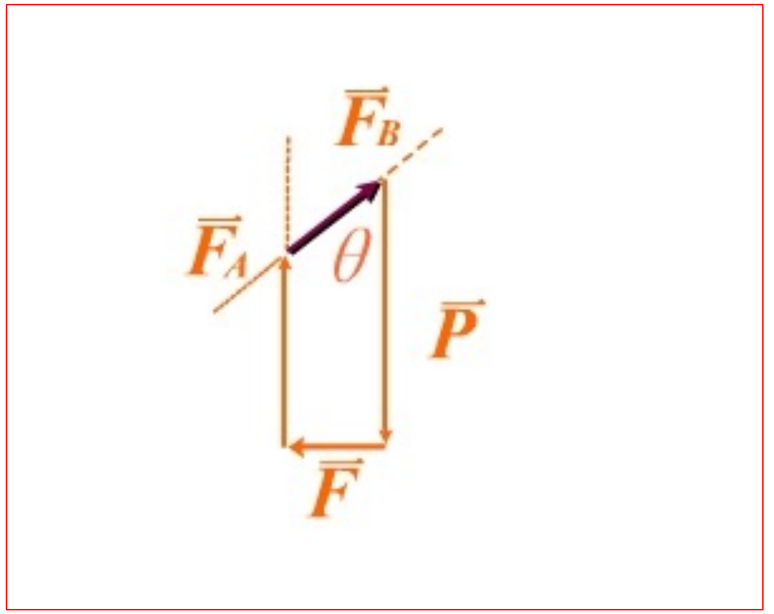
3.What direction is the minimum force magnitude to pull the roller with force, and what is the magnitude of the force?



Solution:

1. Take roller and draw a force diagram Using geometric methods





$$\theta = \arccos\frac{R-h}{R} = 30^0$$

$$F_B \sin\theta = F$$

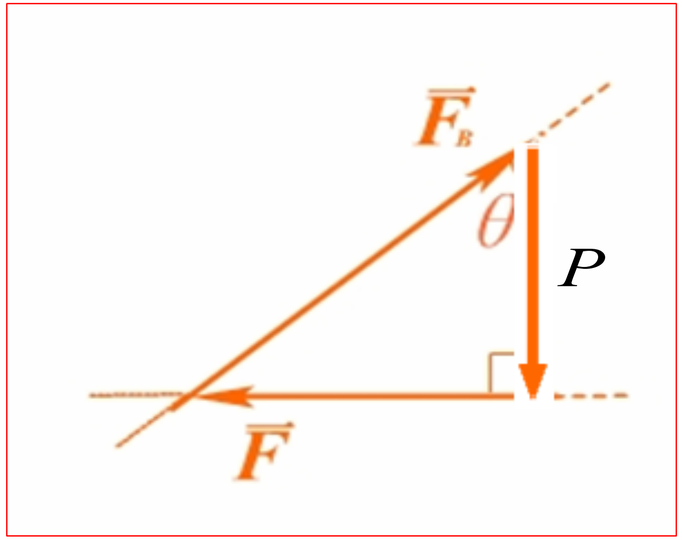
$$F_A + F_B \cos\theta = P$$

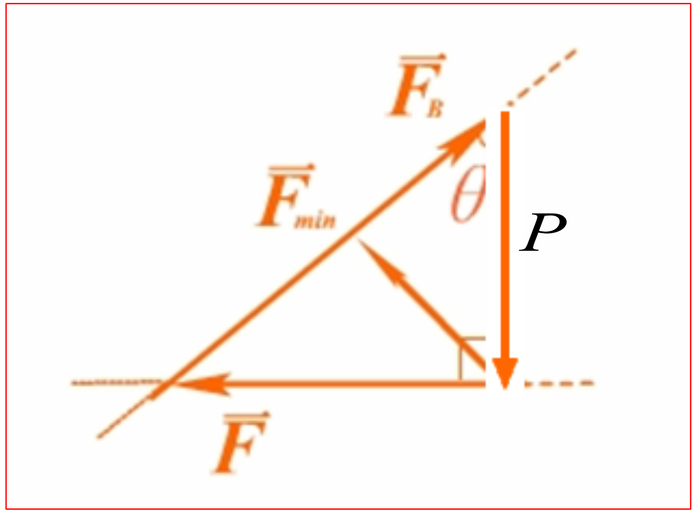
$$F_A = 11.4\text{kN}$$

$$F_B = 10\text{kN}$$

2. Wheels pull over obstacles, $F_A = 0$

Solved using geometric methods $F = P \cdot \tan\theta = 11.55$



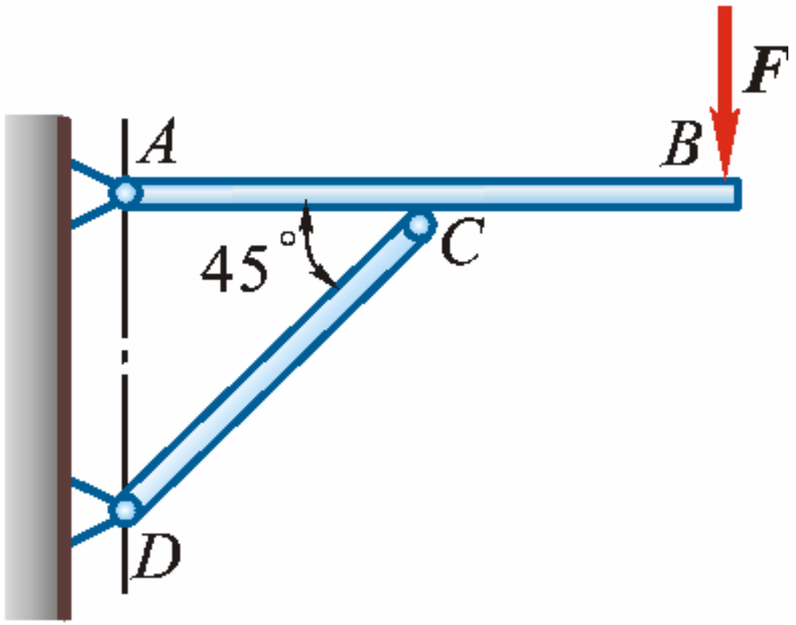


3. Solved $F_{\min} = P \cdot \sin\theta = 10\,\mathrm{kN}$

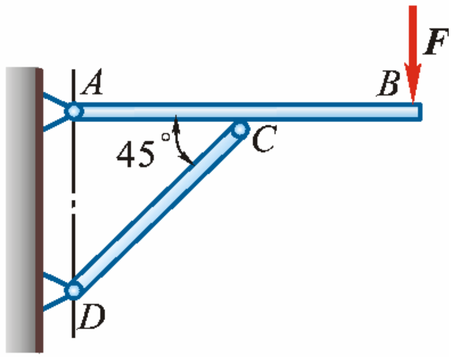
Example 2-2

Known：$AC = CB, F = 10\,\mathrm{kN}$, **ignore the weight of the beam and rod ;**

solve：**The force on rod *CD* and hinge *A* .**



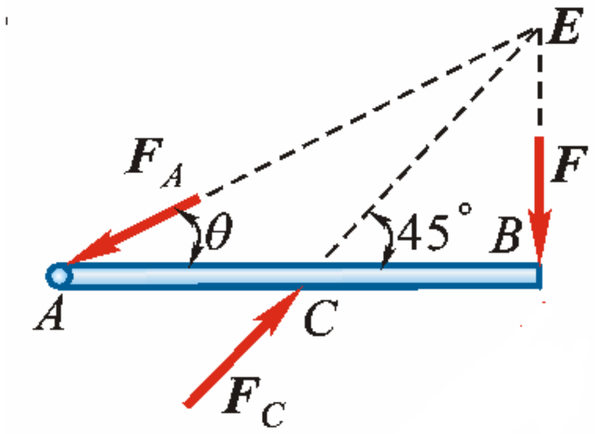
Solution：

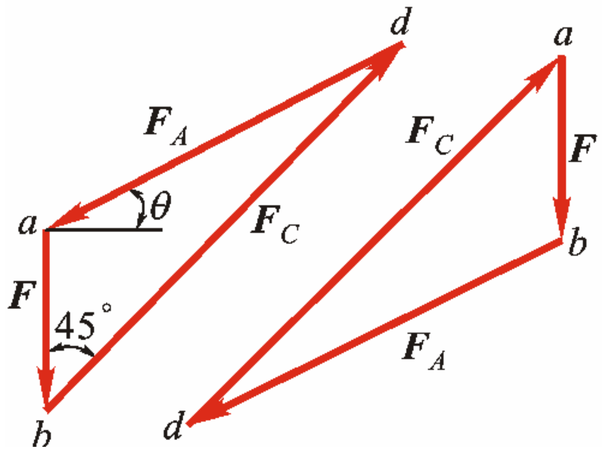


CD is two force rod， take beam *AB*， Draw force diagram.

Using geometric methods, draw a closed force triangle

5 KN





Proportionally measured $F_C = 28.3\text{kN}, F_A = 22.4\text{kN}$

Example
2-3

Known：concurrent force system, $F_1 = 200\text{N}$ $F_2 = 300\text{N}$

$F_3 = 100\text{N}$ $F_4 = 250\text{N}$ **Request: The resultant force**

Solution： **Apply analytical methods**

$$F_{Rx} = \sum F_{ix} = F_1 \cos 30^\circ - F_2 \cos 60^\circ - F_3 \cos 45^\circ + F_4 \cos 45^\circ = 129.3\text{N}$$

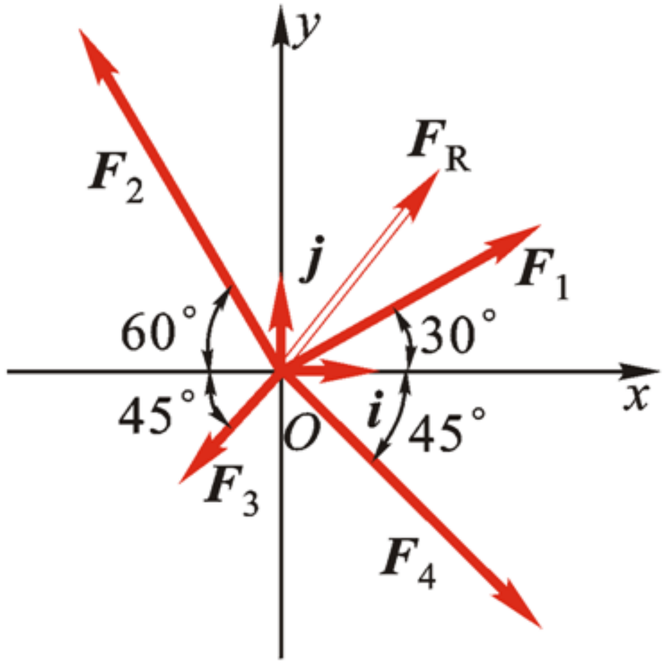
$$F_{Ry} = \sum F_{iy} = F_1 \sin 30^\circ + F_2 \sin 60^\circ - F_3 \sin 45^\circ - F_4 \sin 45^\circ = 112.3\text{N}$$

$$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2} = 171.3\text{N}$$

$$\cos\theta = \frac{F_{Rx}}{F_R} = 0.7548$$

$$\cos\beta = \frac{F_{Ry}}{F_R} = 0.6556$$

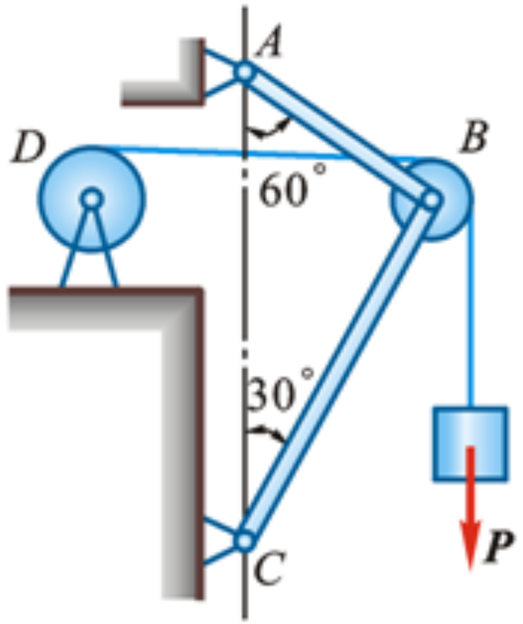
$$\theta = 40.99^\circ, \beta = 49.01^\circ$$



Example 2-4

Known：The force system is shown in the diagram, ignoring the weight of the rods and pulleys, and ignoring the size of the pulleys $P = 20\text{kN}$。

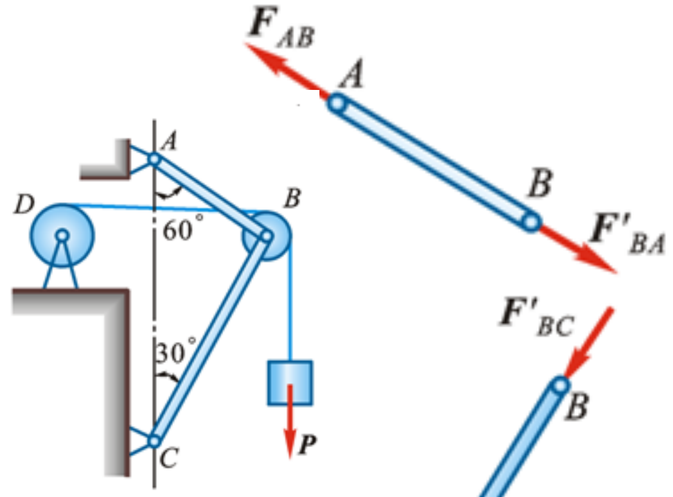
Request: The forces acting on rods AB and BC when the system is in equilibrium.



solution： *AB、BC are two force rods.*

***Pulley B*（or point *B*）**

Force diagram:

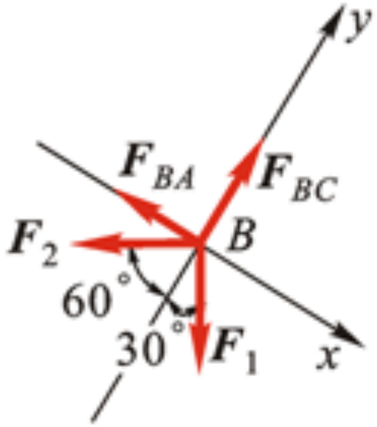


$$\sum F_x = 0 \qquad -F_{BA} + F_1 \cos 60^\circ - F_2 \cos 30^\circ = 0$$

$$\sum F_y = 0 \qquad F_{BC} - F_1 \cos 30^\circ - F_2 \cos 60^\circ = 0$$

$$F_1 = F_2 = P$$

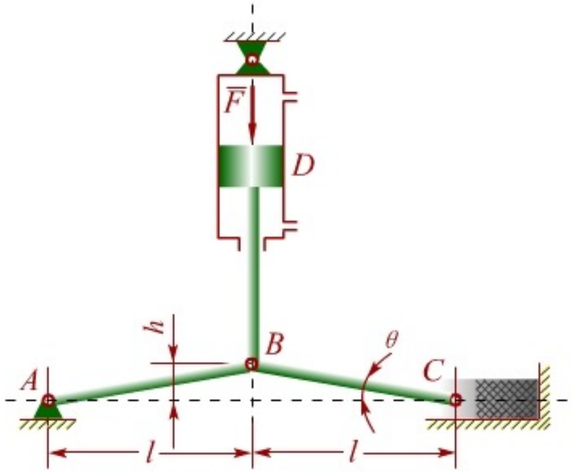
$$F_{BA} = -7.321\text{kN} \quad F_{BC} = 27.32\text{kN}$$



Example 2-5

Known： F=3kN, l=1500mm, h=200mm, **Ignore self weight；**

Request: When in equilibrium, the pressure of block ***C*** on the workpiece and the ground, and the force on ***AB*** rod



Solution： ***AB、BC are two force rods.***

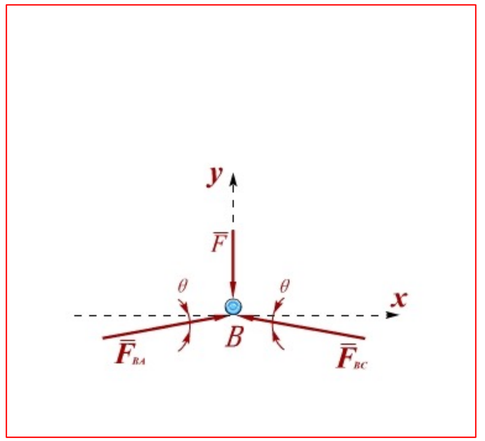
Pin *B*:

$$\sum F_x = 0 \quad F_{BA}\cos\theta - F_{BC}\cos\theta = 0$$

$$\Rightarrow \quad F_{BA} = F_{BC}$$

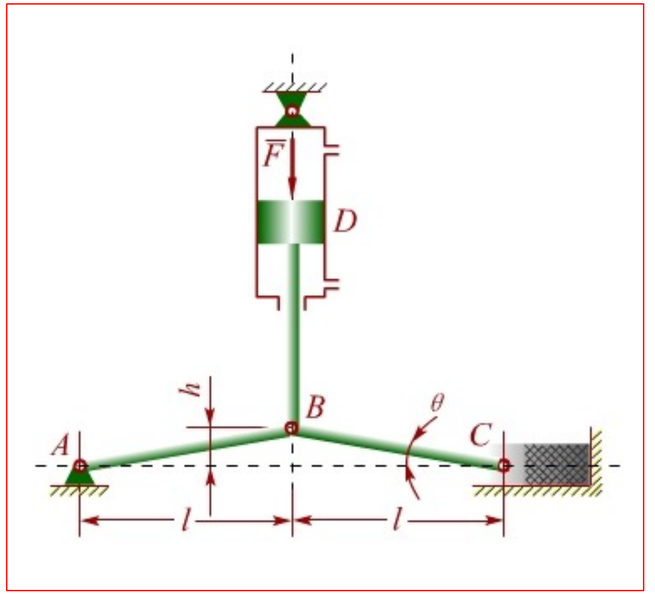
$$\sum F_y = 0 \quad F_{BA}\sin\theta + F_{BC}\sin\theta - F = 0$$

$$\Rightarrow \quad F_{BA} = F_{BC} = 11.35\text{kN}$$



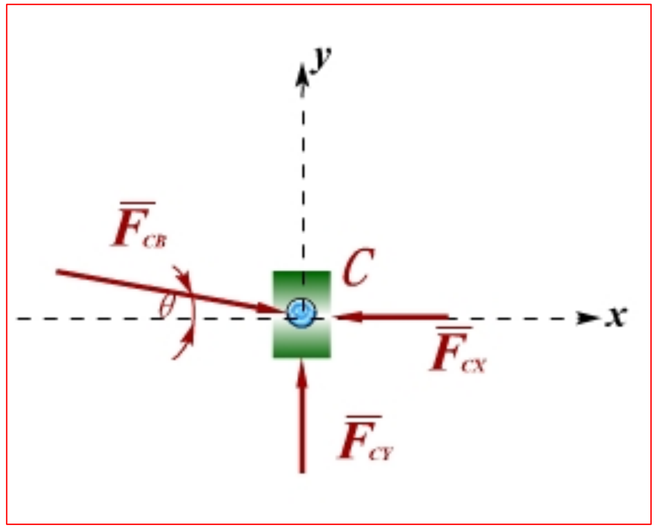
Block *C*:

$$\sum F_x = 0 \qquad F_{CB}\cos\theta - F_{Cx} = 0$$



$$F_{Cx} = \frac{F}{2}\cot\theta = \frac{Fl}{2h} = 11.25\text{kN}$$

$$\sum F_y = 0 \qquad -F_{CB}\sin\theta + F_{Cy} = 0$$



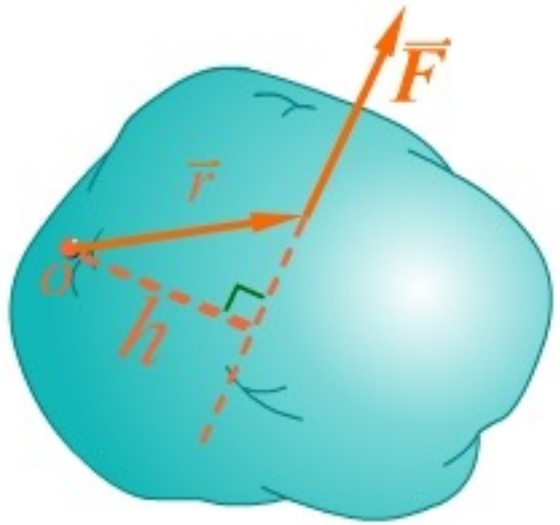
$$F_{Cy} = 1.5\text{kN}$$

Chapter 2. Planar force system

- 2.1 Coplanar concurrent force system
- **2.2 Moment of a planar force about a point**
- 2.3 The equilibrium conditions and equilibrium equations for a planar arbitrary force system
- 2.4 Internal Force Calculation of Simple Plane Trusses

• 2.2 Moment of a planar force about a point

I、The Moment of a Planar Force About a Point



On Moment acting surface:

O **is called the center of the moment (centroid),** ***r*** **is called radius vector;**

The vertical distance ***h*** **from** ***O*** **to the line of action of force is called the force arm**

The moment of force:

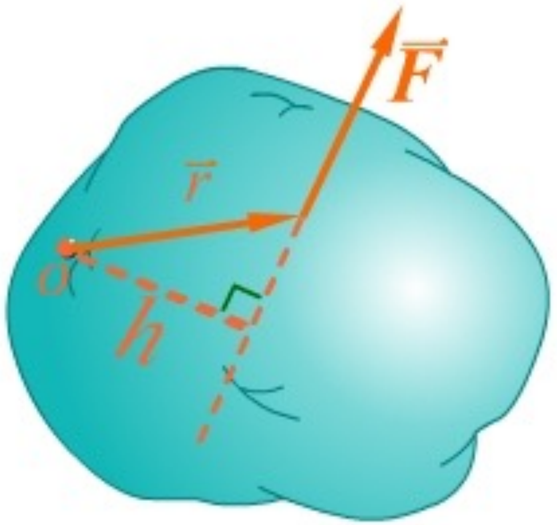
The effect of force on an object rotating around a centroid.

$$Mo(F) = F \cdot r$$

The unit of moment is: N·m or KN·m

• 2.2 Moment of a planar force about a point

I、The Moment of a Planar Force About a Point



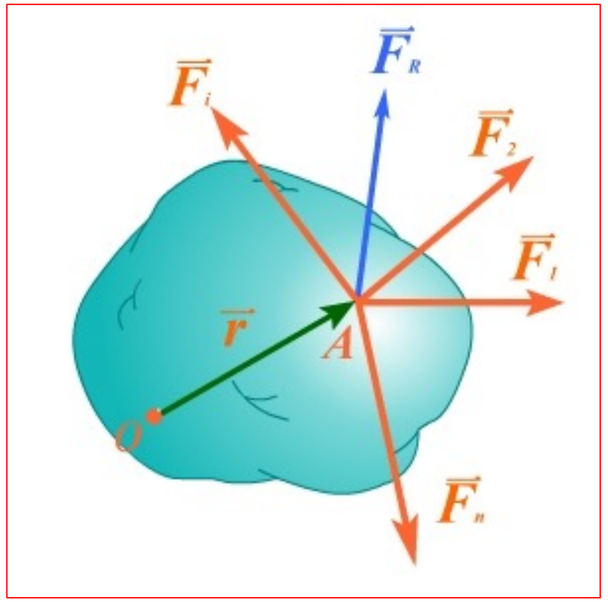
Two elements of the moment：

1. Moment magnitude: product of force F and force arm

2. Moment direction: Rotation direction of the moment

The moment of force on a point is an algebraic quantity, whose absolute value is equal to the product of the magnitude of the force and the arm of the force. Its sign is positive (+) when the force causes an object to turn counterclockwise around the center of the moment, and negative (-) when it doesn't.

II、 The resultant moment theorem and analytical expression of moment



The resultant moment theorem: The moment of any point in the plane is equal to the algebraic sum of the moments of all force components at that point.

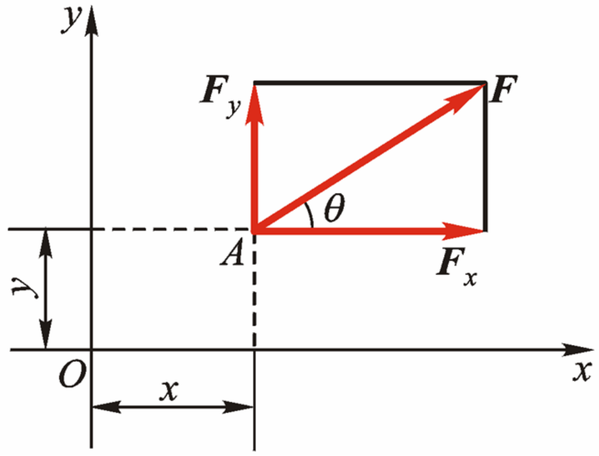
$$M_O(F_R) = \sum_{i=1}^{n} M_o(F_i)$$

$$M_O(F) = M_o(F_y) - M_o(F_x)$$

$$= x \cdot F \cdot \sin\theta - y \cdot F \cdot \cos\theta$$

$$= xF_y - yF_x$$

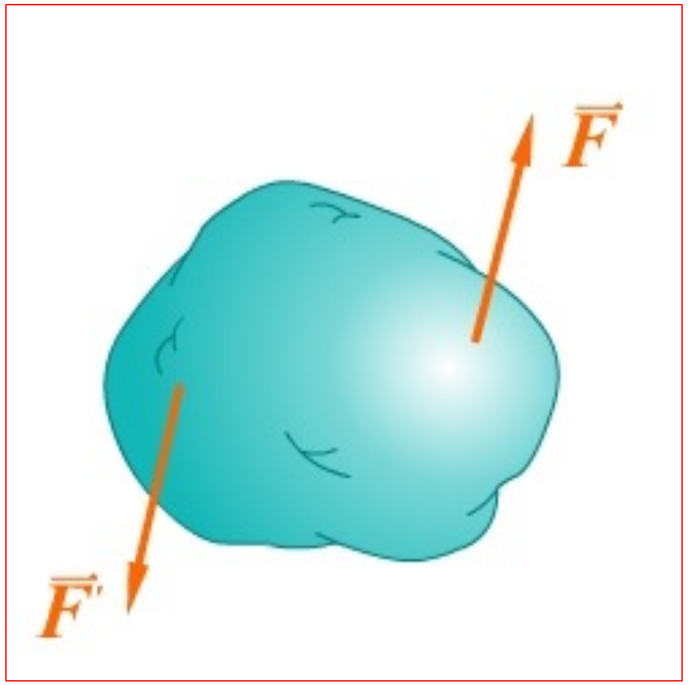
$$M_O(F_R) = \Sigma x_i(F_{iy}) - \Sigma y_i(F_{ix})$$

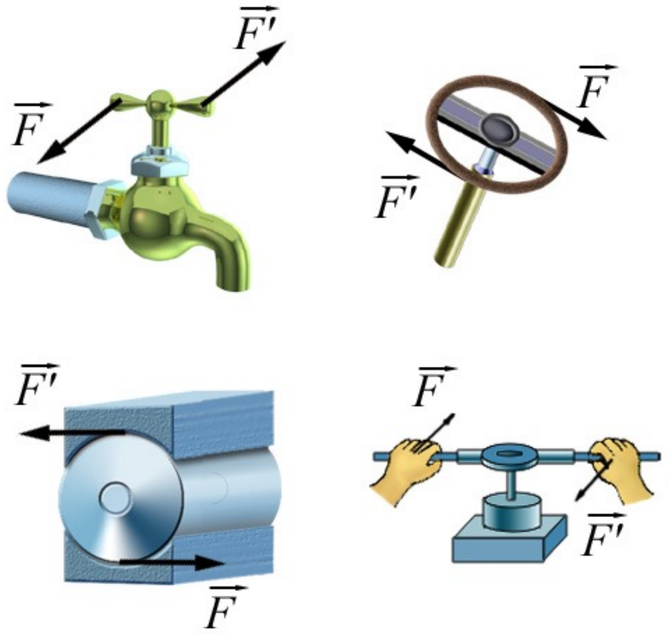


III、 Force couple and moment of force couple

Force couple

A force system composed of two equal, opposite, and non collinear (parallel) forces is called a couple, denoted as $(F\ ,F')$

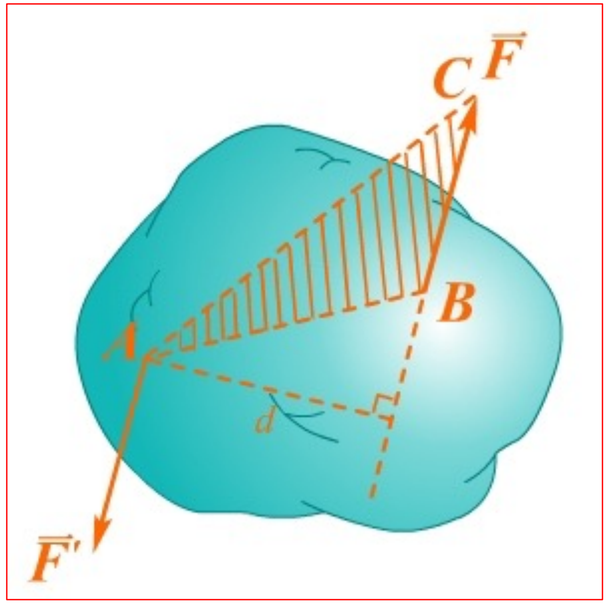




Moment of force couple

The plane where the two forces in a couple are located is called the plane of action of the couple

The vertical distance between two forces in a force couple is called a force couple arm



Two elements:

A. Magnitude: product of force and couple arm

B. Direction: Rotation direction

Moment of force couple

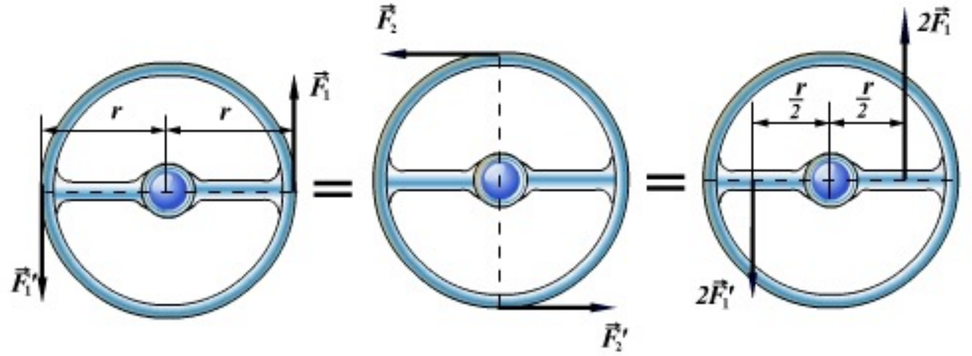
$$M = \pm F \cdot d = \pm 2\Delta ABC$$

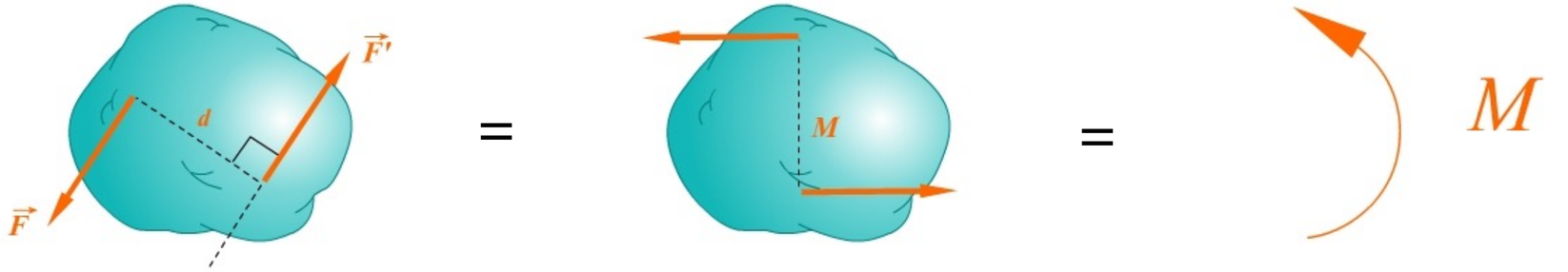
IV、The Equivalent Theorem of Coplanar Force Couple

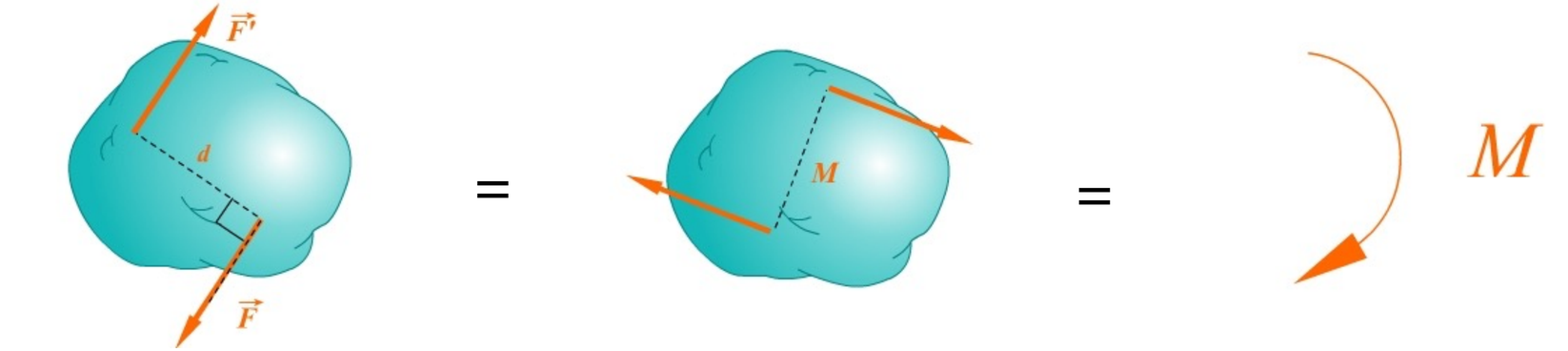
Theorem：If two force couples in the same plane have equal moments, then they are equivalent to each other.

corollary

- **Any force couple can transfer freely within its surface of action without changing its effect on the rigid body. Therefore, the effect of a couple on a rigid body is independent of the position of the couple within its plane of action.**
- **As long as the moment of the couple remains constant, the magnitude of the force in the couple and the force can be changed simultaneously, and the effect on the rigid body remains unchanged.**
- **The arms of a force couple and the magnitude of the force are not characteristic quantities of the force couple, only the moment of the force couple is the only measure of the action of a planar force couple.**


$\vec{F}_2$
$2\vec{F}_1$
$\vec{F}_1$
r
r
$\frac{r}{2}$
$\frac{r}{2}$
$\vec{F}_1'$
$2\vec{F}_1'$
$\vec{F}_2'$


$\vec{F}'$
d
$\vec{F}$
=
M
=
M


$\vec{F}'$
d
$\vec{F}$
=
M
=
M

V、Composition and equilibrium conditions of planar force couple system

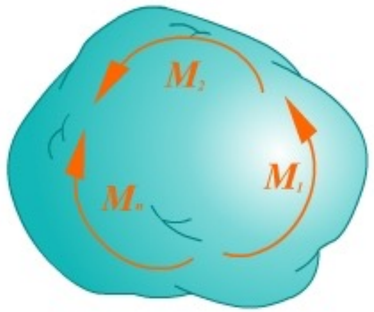
Known： $M_1, M_2, \ldots\ldots, M_n$

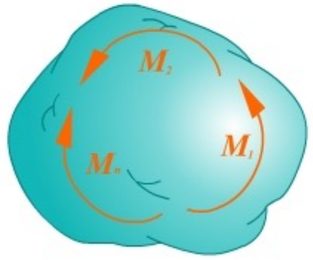
Choose any distance *d*

$$\frac{M_1}{d} = F_1 \qquad M_1 = F_1 d$$

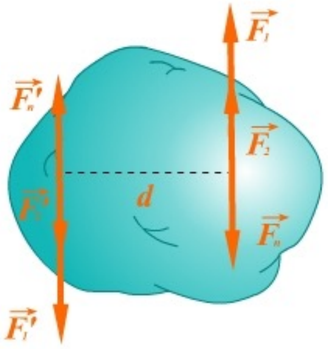
$$\frac{M_2}{d} = F_2 \qquad M_2 = F_2 d$$

$$\left|\frac{M_n}{d}\right| = F_n \qquad M_n = -F_n d$$

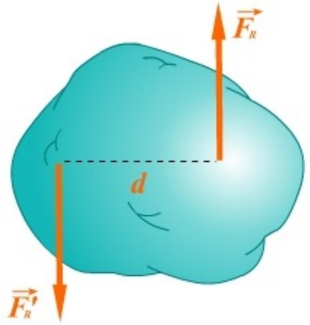




=

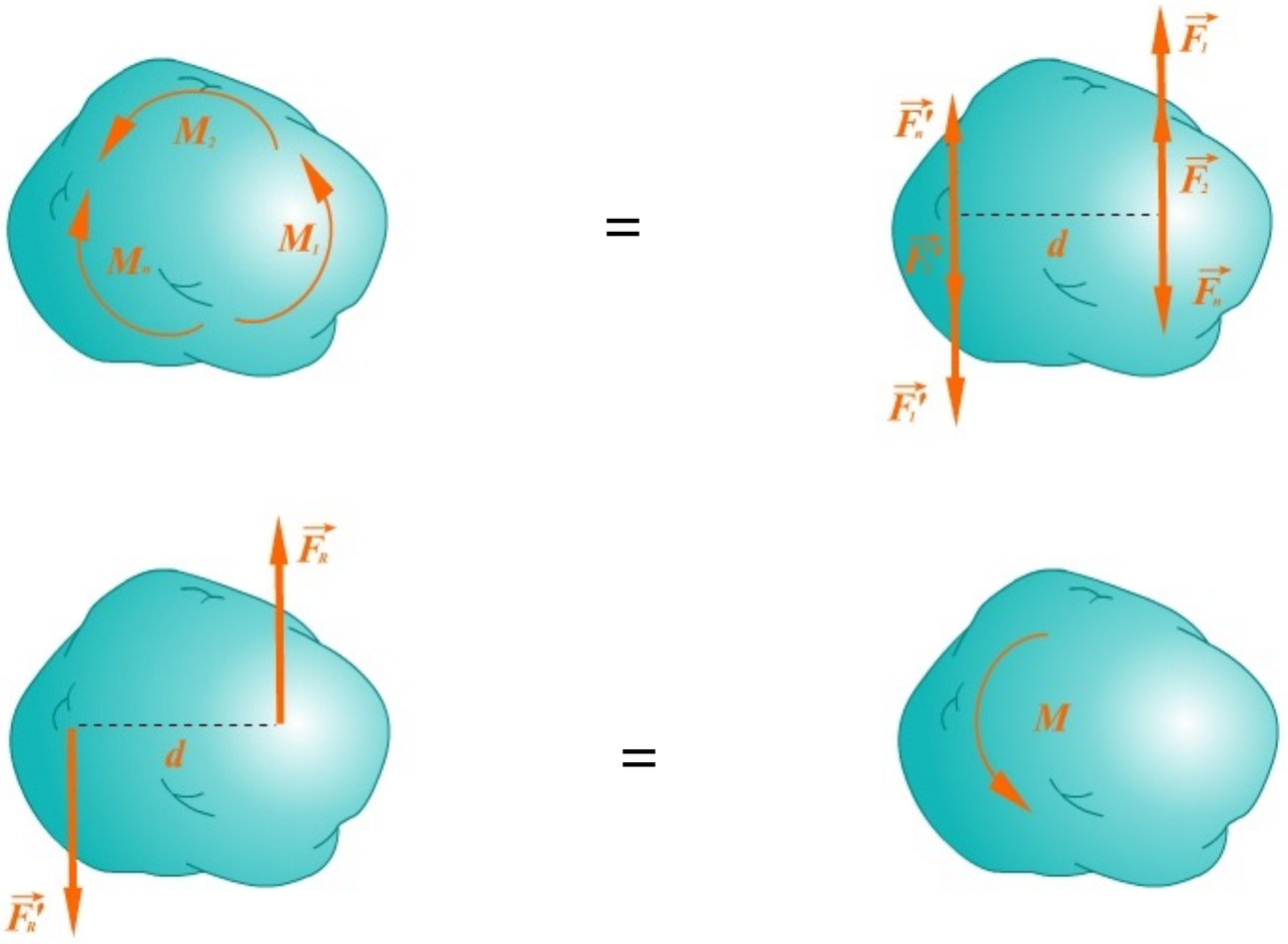


=



$$F_R = F_1 + F_2 + \ldots\ldots - F_n$$

$$F'_R = F'_1 + F'_2 + \ldots\ldots - F'_n$$



$$M = F_R\ d = F_1 d + F_2 d + \ldots\ldots - F_n d = M_1 + M_2 \ldots\ldots + M_n$$

$$M = \sum_{i=1}^{n} M_i = \sum M_i$$

The Necessary and Sufficient Conditions for Equilibrium of Plane Force Couple System $M = 0$, There is the following equilibrium equation

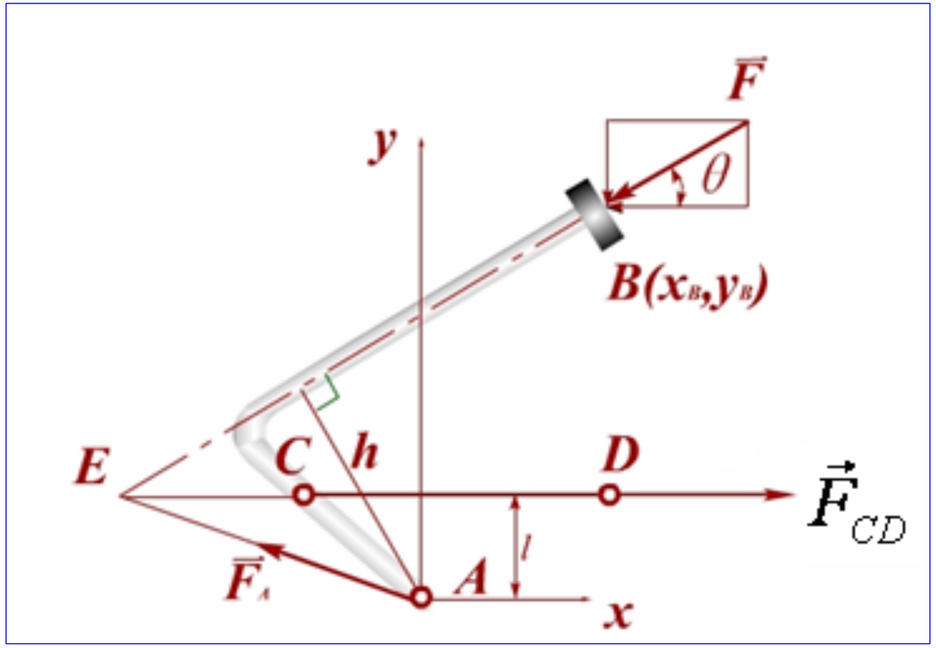
$$\sum M_i = 0$$

The necessary and sufficient condition for the equilibrium of a planar force couple system is that the algebraic sum of all force couple moments is equal to zero

Example 2-6

Known: $F, \theta, x_B, y_B, l;$

Solve: **The tension of the *CD* rod in equilibrium.**



Solution: CD **is a two force rod, take the pedal**

Take the moment of point A based on the equilibrium condition

$$\Sigma M_A=0 \qquad F\cos\theta \cdot y_B - F\sin\theta \cdot x_B - F_{CD} \cdot l = 0$$

Solved: $$F_{CD} = \frac{F\cos\theta \cdot y_B - F\sin\theta \cdot x_B}{l}$$

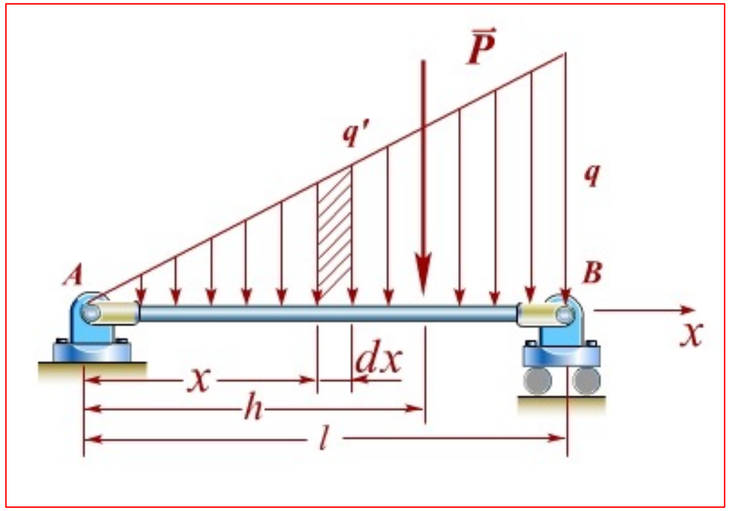
Example 2-7: Known q, l;

Request： The equivalent concentrated force and the line of action of the force

Solution： Take the element of uniformly distributed load

$$q' = \frac{x}{l} \cdot q$$

$$P = \int_0^l \frac{x}{l} \cdot q \cdot \mathrm{d}x = \frac{1}{2} ql$$



According to the law of resultant moment

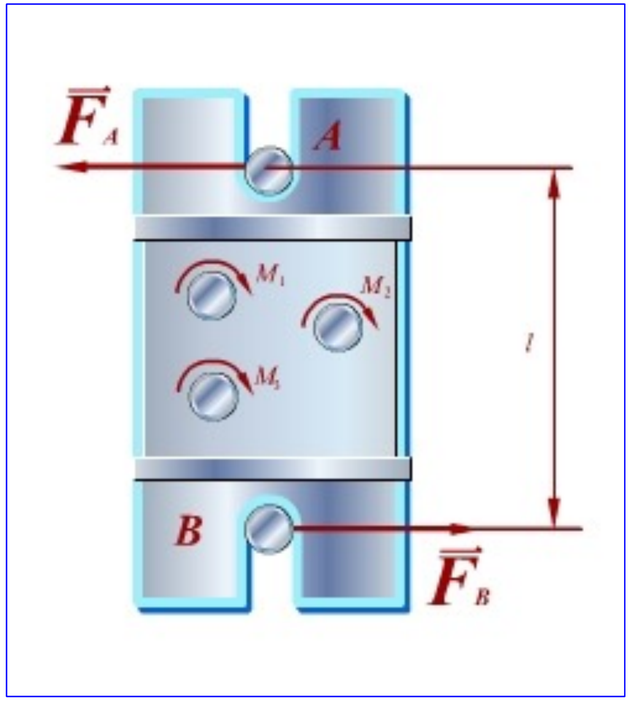
$$P \cdot h = \int_0^l q' \cdot \mathrm{d}x \cdot x = \int_0^l \frac{x^2}{l} q \cdot \mathrm{d}x$$

Solve: $h = \frac{2}{3} l$

Example 2-8

Known: $M_1 = M_2 = 10\mathbf{N}\cdot\mathbf{m}, M_3 = 20\mathbf{N}\cdot\mathbf{m}, l = 200\mathbf{mm};$

Request: The horizontal force exerted on the smooth bolt AB



solution: **According the equilibrium equation of resultant moment:**

$$\sum M = 0$$

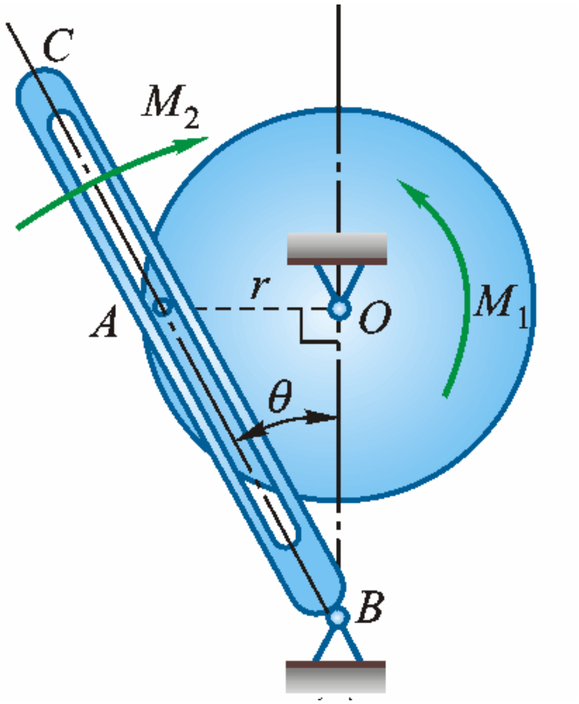
$$F_A l - M_1 - M_2 - M_3 = 0$$

Solve: $$F_A = F_B = \frac{M_1 + M_2 + M_3}{l} = 200\mathbf{N}$$

Example **2-9**

Known: $M_1 = 2\text{kN}\cdot\text{m}, OA = r = 0.5\text{m}, \theta = 30^0$

Request: The moment M_2 at equilibrium and the constraint force at hinge O, B

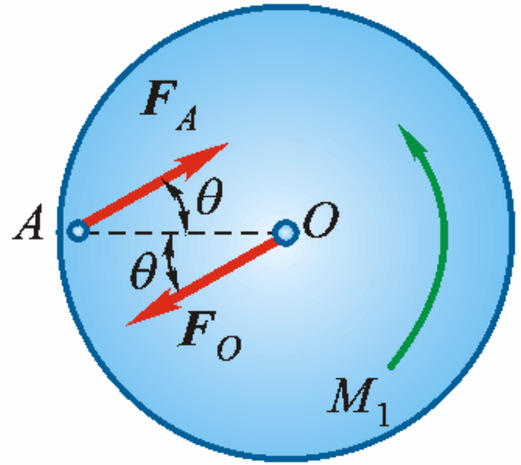


solution：

Take the wheel based on the property that the force couple can only be balanced by the force couple

$$\sum M = 0 \qquad M_1 - F_A \cdot r\sin\theta = 0$$

Solve: $F_O = F_A = 8\text{kN}$

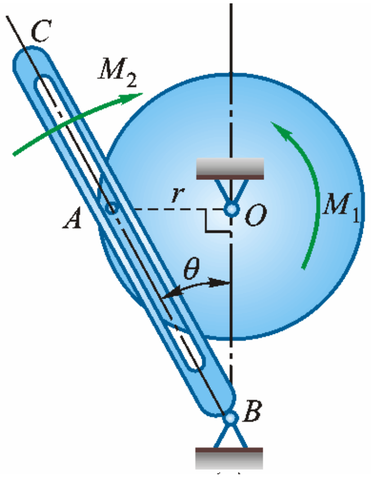


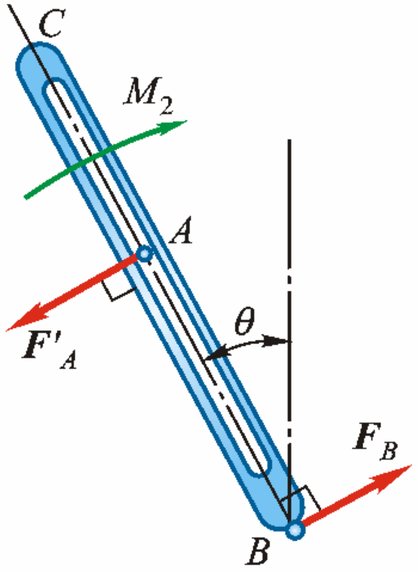
Take BC , **draw force diagram**

$$\sum M = 0 \qquad F_A' \cdot \frac{r}{\sin\theta} - M_2 = 0$$

Solve: $M_2 = 8\text{kN}\cdot\text{m}$

$F_B = F_A = 8\text{kN}$





Chapter 2. Planar force system

- 2.1 Coplanar concurrent force system
- 2.2 Moment of a planar force about a point
- **2.3 The equilibrium conditions and equilibrium equations for a planar arbitrary force system**
- 2.4 Internal Force Calculation of Simple Plane Trusses

• 2.3 The equilibrium conditions and equilibrium equations for a planar arbitrary force system

I. The equilibrium equation of arbitrary planar force system

The necessary and sufficient conditions for equilibrium of any planar force system are:

$$F_R = 0 \qquad M_O = 0$$

Since

$$F_R = \sqrt{F_{Rx}^{\ 2} + F_{Ry}^{\ 2}} \qquad M = \sum_{i=1}^{n} M_i = \sum M_i$$

The equilibrium equation of arbitrary planar force system

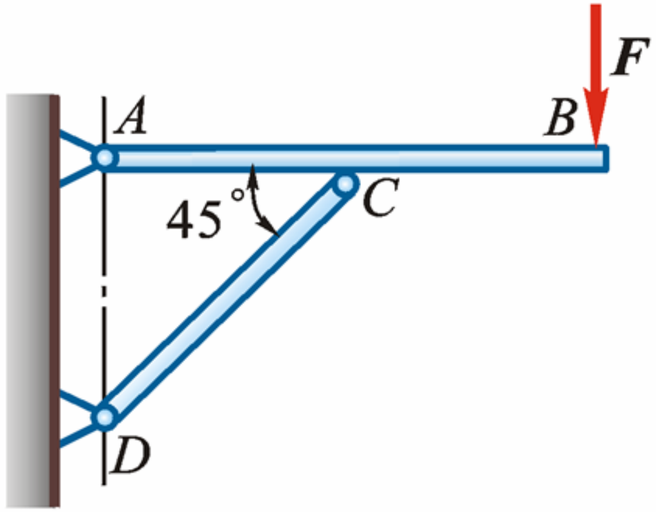
$$\begin{cases} \sum F_x = 0 \\ \sum F_y = 0 \\ \sum M_O = 0 \end{cases}$$

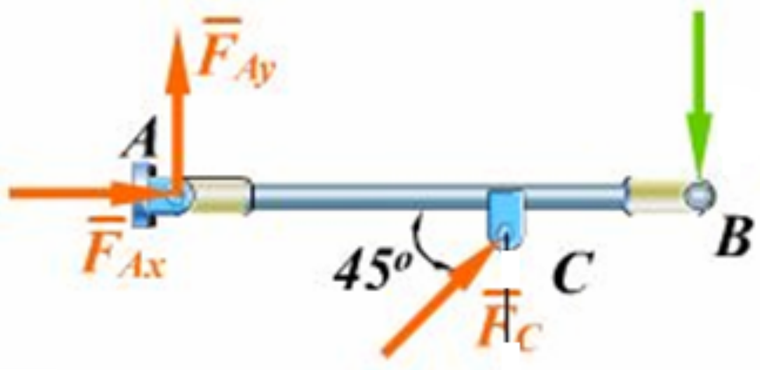
The analytical condition for equilibrium of any planar force system is:
The algebraic sum of the projections of all forces on two optional coordinate axes is equal to zero, and the algebraic sum of the moments of each force at any point is also equal to zero

Example **2-10**

Known：$AC = CB = l, F = 10\,\mathrm{kN}$

Request：**The force on rod *CD* and hinge *A* .**





Solution：**Take AB, draw force diagram**

$$\sum F_x = 0 \qquad F_{Ax} + F_C \cos 45^0 = 0$$

$$\sum F_y = 0 \qquad F_{Ay} + F_C \sin 45^0 - F = 0$$

$$\sum M_A = 0 \qquad F_C \cos 45^0 \cdot l - F \cdot 2l = 0$$

$$F_C = 28.28\mathrm{kN}, F_{Ax} = -20\mathrm{kN}, F_{Ay} = -10\mathrm{kN}$$

Example **2-11**

Known：$P_1 = 10\text{kN},\quad P_2 = 40\text{kN},$ **Demensions show in the figure**

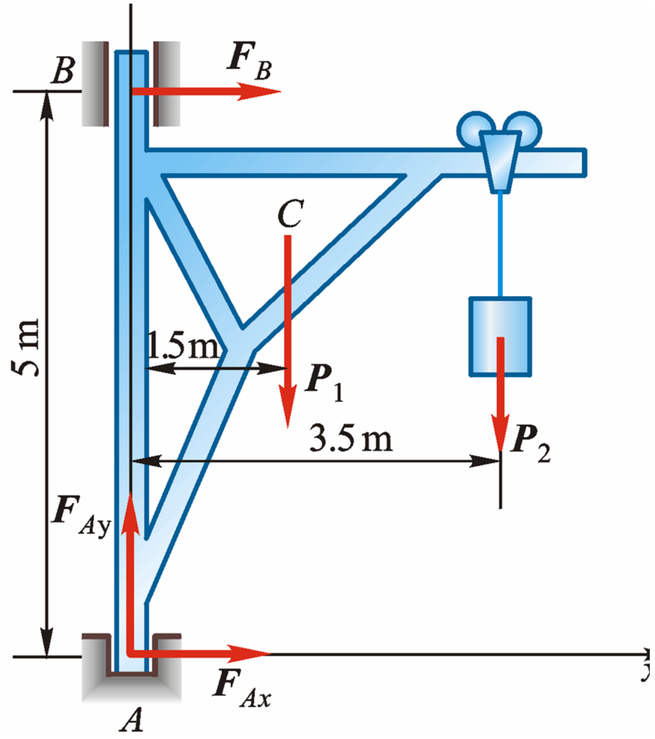
Request： **The constraint force at the bearing** A, B

Solution： **Take the crane and draw the force diagram**

$$\sum F_x = 0 \qquad F_{Ax} + F_B = 0$$

$$\sum F_y = 0 \qquad F_{Ay} - P_1 - P_2 = 0$$

$$\sum M_A = 0 \qquad -F_B \cdot 5 - 1.5 \cdot P_1 - 3.5 \cdot P_2 = 0$$

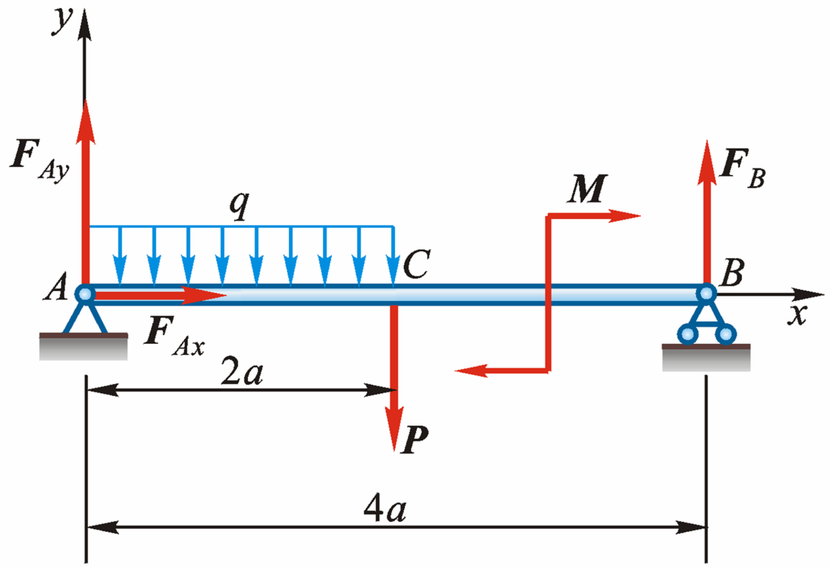


$$F_{Ay} = 50\text{kN} \qquad F_B = -31\text{kN} \qquad F_{Ax} = 31\text{kN}$$

Example **2-12**

Known: $P,\ q,\ a,\ M = Pa$

Request: **Constraint force at hinge support** A, B



Solution： **Take beam** AB **, draw force diagram.**

$$\sum F_x = 0 \qquad F_{Ax} = 0$$

$$\sum M_A = 0 \quad F_B \cdot 4a - M - P \cdot 2a - q \cdot 2a \cdot a = 0$$

$$F_B = \frac{3}{4}P + \frac{1}{2}qa$$

$$\sum F_y = 0 \qquad F_{Ay} - q \cdot 2a - P + F_B = 0$$

$$F_{Ay} = \frac{P}{4} + \frac{3}{2}qa$$

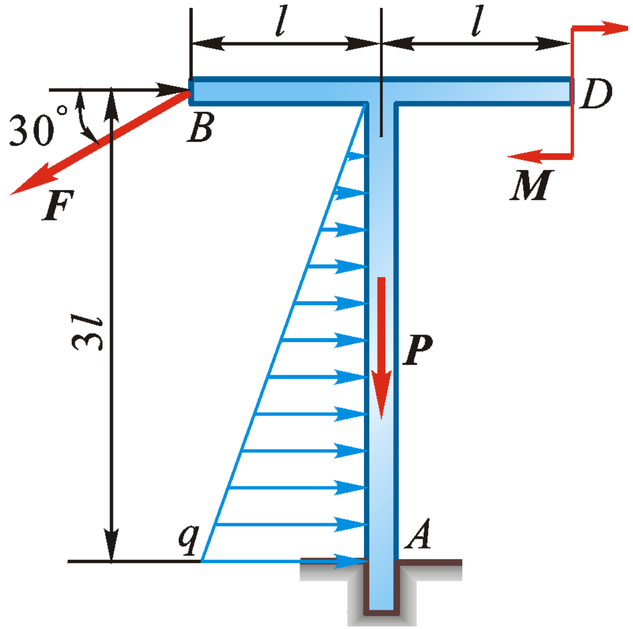
example 2-13

Known: $P = 100\text{kN}, M = 20\text{kN}\cdot\text{m},$

$q = 20\,\text{kN/m}, F = 400\text{kN}, l = 1\text{m}$

Request: **constraint force at Fixed end** A

Solution: Take the T-shaped frame and draw a force diagram



In which $F_1 = \frac{1}{2} q \times 3l = 30\text{kN}$

$$\sum F_x = 0 \qquad F_{Ax} + F_1 - F\sin 60^0 = 0$$

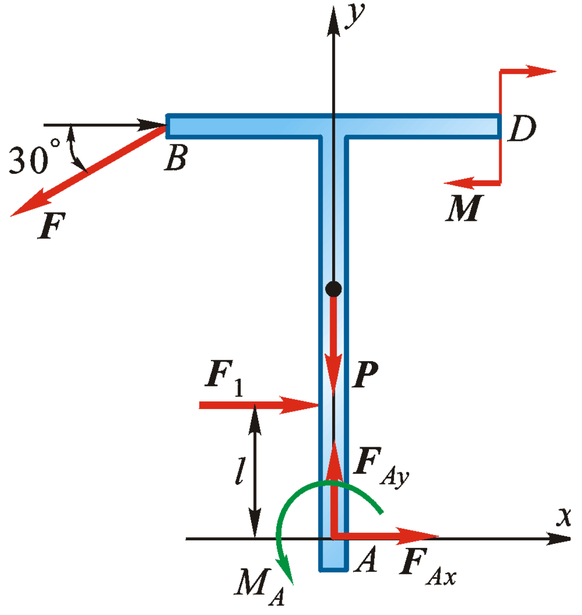
$$\sum F_y = 0 \qquad F_{Ay} - P - F\cos 60^0 = 0$$

$$\sum M_A = 0$$

$$M_A - M - F_1 \cdot l + F\cos 60^0 \cdot l + F \cdot \sin 60^0 \cdot 3l = 0$$

$$\Rightarrow \quad F_{Ax} = 316.4\text{kN} \qquad F_{Ay} = 300\text{kN}$$

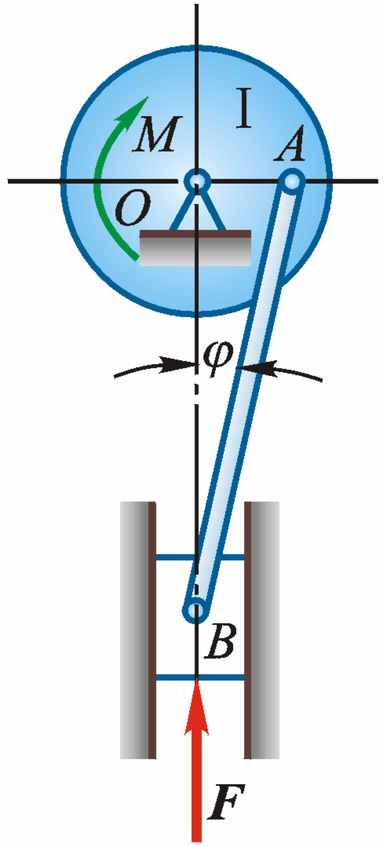
$$M_A = -1188\text{kN}\cdot\text{m}$$



Example 2-14 Known: F, $OA = R, AB = l,$

Ignore self weight and friction of the components, system is equilibrium; Request:

The magnitude of the **coupling moment** M, the restraining **force** at the bearing O, the force on the connecting **rod** AB, and the lateral force exerted by the punch on the guide rail

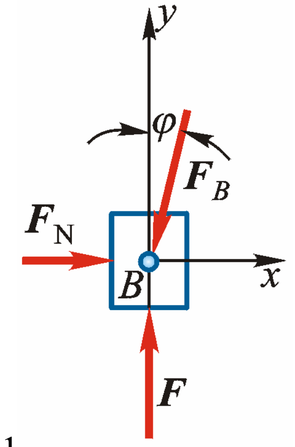


Solution: Take punch B and draw force diagram:

$$\sum F_y = 0 \qquad F - F_B \cos\varphi = 0$$

$$\sum F_x = 0 \qquad F_N - F_B \sin\varphi = 0$$

$$F_B = \frac{F}{\cos\varphi} = \frac{Fl}{\sqrt{l^2 - R^2}} \qquad F_N = F\tan\varphi = \frac{FR}{\sqrt{l^2 - R^2}}$$



Take the wheel, draw force diagram:

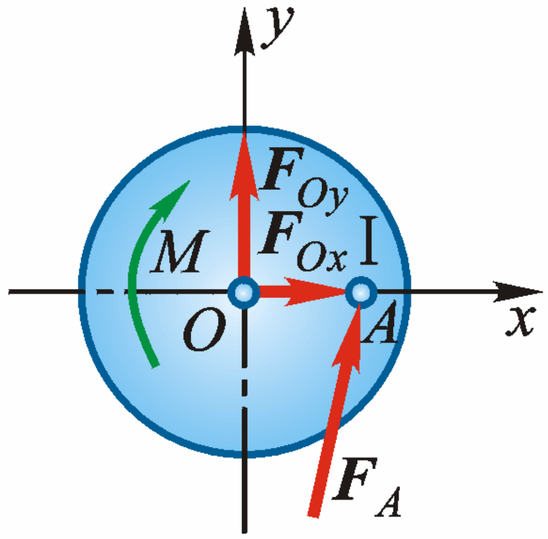
$$\sum F_x = 0 \qquad F_{Ox} + F_A \cos\varphi = 0$$

$$\sum F_y = 0 \qquad F_{Oy} + F_A \sin\varphi = 0$$

$$\sum M_O = 0 \qquad F_A R \cos\varphi - M = 0$$

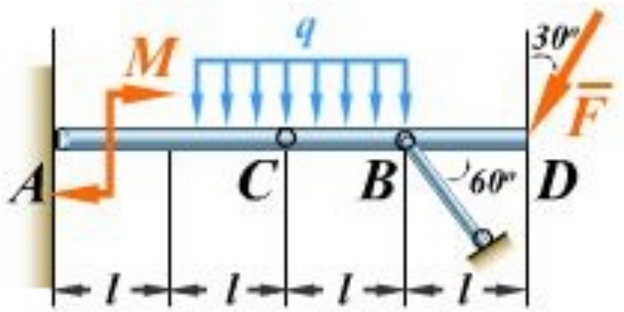
$$F_{Ox} = -\frac{FR}{\sqrt{l^2 - R^2}} \qquad F_{Oy} = -F$$

$$M = FR$$

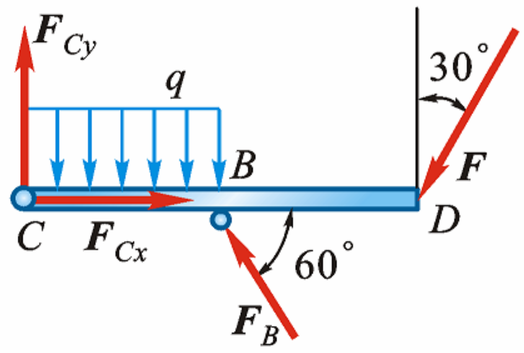


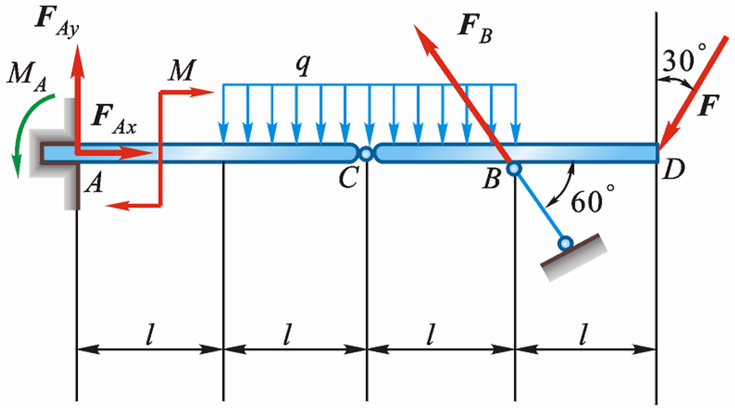
Example 2-15 **Known** F=20kN, q=10kN/m, M = 20kN·m, l=1m;

Request: Constraint force at ***A, and B*** . （**A is fixed end**）



Solution **:** **Take beam *CD* （*isolated body*）, draw force diagram:**





$$\sum M_C = 0$$

$$F_B \sin 60^0 \cdot l - -ql \cdot \frac{l}{2} - F\cos 30^0 \cdot 2l = 0$$

$$\Rightarrow \quad F_B = 45.77\text{kN}$$

Take the entire system, draw free-body diagram

$$\sum F_x = 0$$

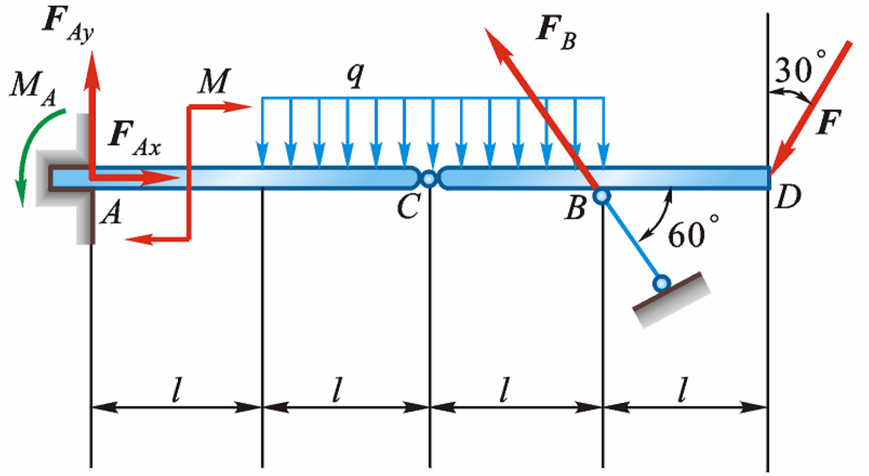
$$F_{AX} - F_B \cos 60^0 - F \sin 30^0 = 0$$

$$\sum F_y = 0$$

$$F_{AY} + F_B \sin 60^0 - F \cos 30^0 = 0$$

$$\sum M_A = 0$$

$$M_A - M - 2ql \cdot 2l + F_B \sin 60^0 \cdot 3l - F \cos 30^0 \cdot 4l = 0$$

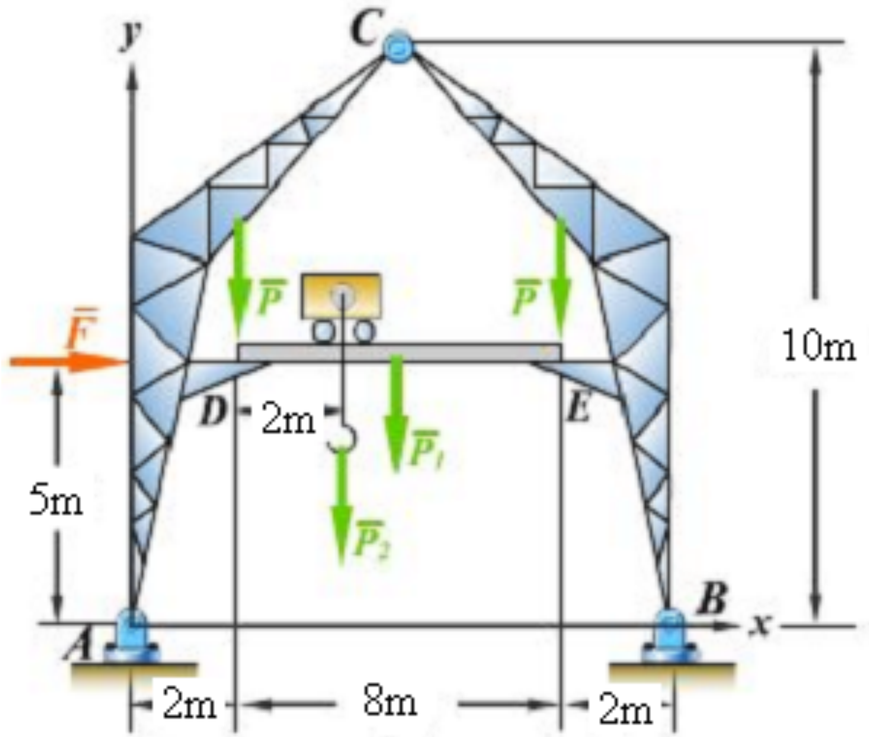


Are the equilibrium equations correct?

$$\Rightarrow \quad M_A = 10.37\text{kN} \cdot \text{m} \qquad F_{Ax} = 32.89\text{kN} \qquad F_{Ay} = -2.32\text{kN}$$

Example 2-16

Given：P=60kN, P_1=20kN, P_2=10kN, wind load F=10kN, with dimensions as shown in the diagram. Calculate the constrain forces at A,B



Solution

Take the entire system, draw free-body diagram

$$\sum M_A = 0 \quad 12F_{By} - 10P - 6P_1 - 4P_2 - 2P - 5F = 0$$

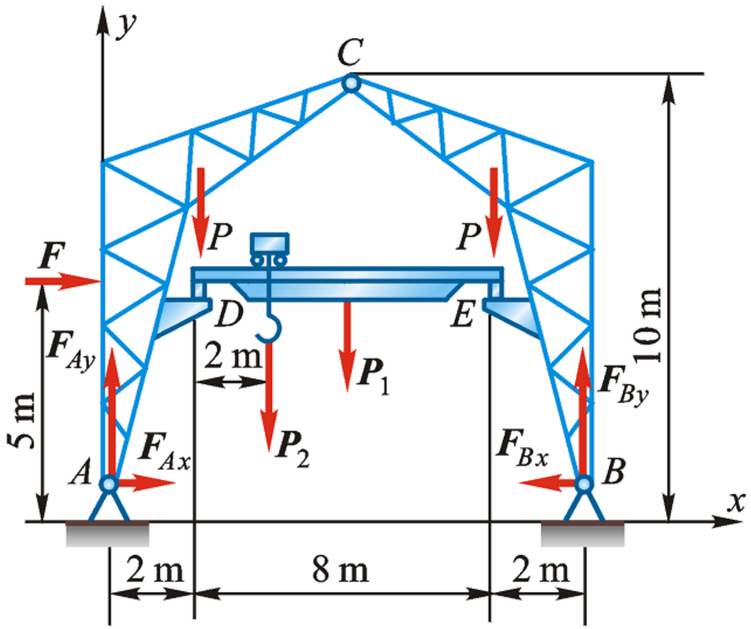
$$\sum F_y = 0 \quad F_{Ay} + F_{By} - 2P - P_1 - P_2 = 0$$

$$\sum F_x = 0 \quad F_{Ax} + F - F_{Bx} = 0$$

$$F_{Ay} = 72.5\text{kN}$$

$$F_{By} = 77.5\text{kN}$$

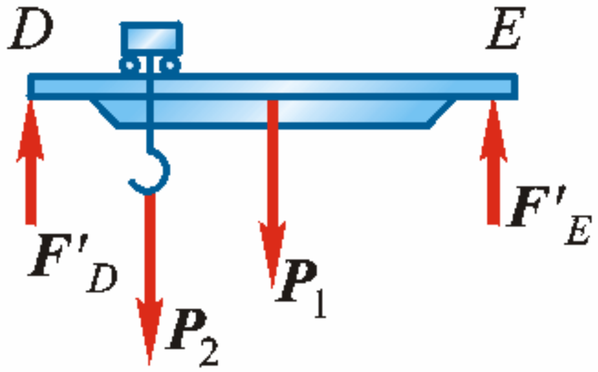
$$F_{Ax} = F_{Bx} - F$$



Take the crane beam *DE* and draw the force diagram

$$\sum M_D = 0 \qquad 8F_E^{'} - 4P_1 - 2P_2 = 0$$

$$\Rightarrow \qquad F_E^{'} = 12.5\text{kN}$$



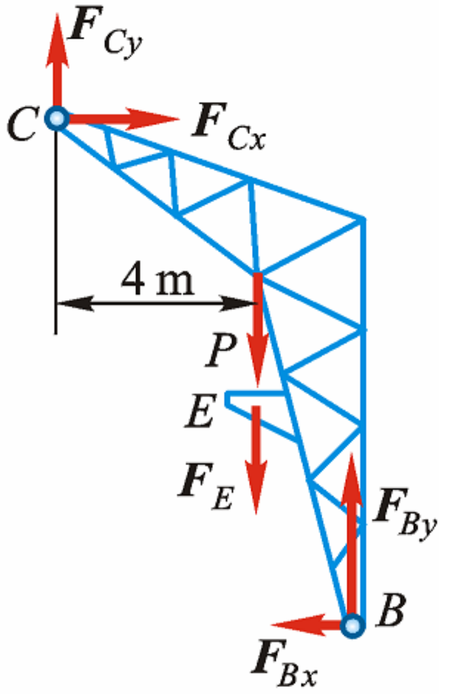
Take the right component and draw a force diagram

$$\sum M_C = 0$$

$$6F_{By} - 10F_{Bx} - 4\left(P + F_E\right) = 0$$

$$\Rightarrow \qquad F_{Bx} = 17.5\text{kN}$$

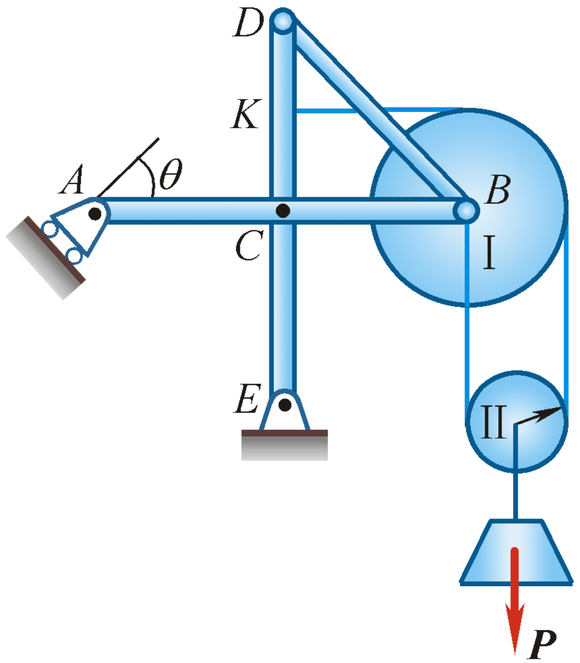
$$F_{Ax} = 7.5\text{kN}$$



Example 2-17

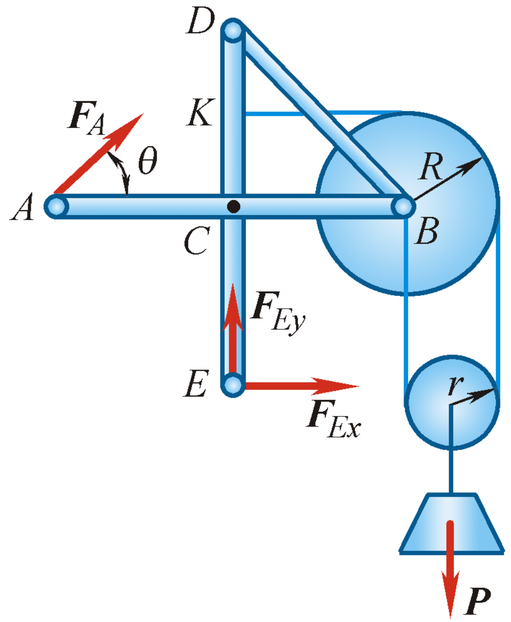
Known: $DC=CE=CA=CB=2l$, $R=2r=l$, P , **disregarding the self weight of each component,** $\theta = 45^0$.

Request: constraint force at hinge supports A and E and force on rod BD



Solution:

Take the entire system, draw free-body force diagram



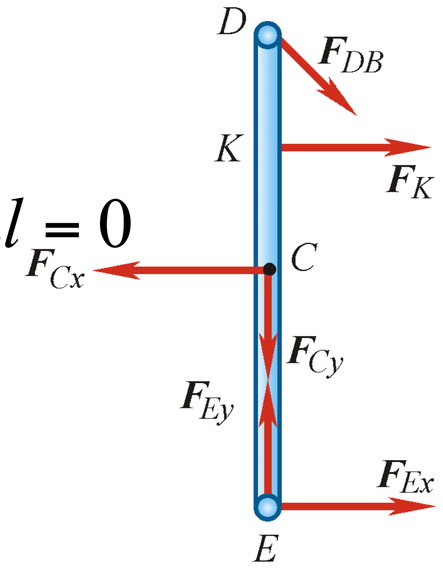
$$\sum M_E = 0 \quad -F_A \cdot \sqrt{2} \cdot 2l - P \cdot \frac{5}{2} l = 0$$

$$\sum F_x = 0 \quad F_{Ex} + F_A \cos 45^0 = 0$$

$$\sum F_y = 0 \quad F_{Ey} - P + F_A \sin 45^0 = 0$$

$$\Rightarrow \quad F_A = -\frac{5\sqrt{2}}{8} P \quad F_{Ex} = \frac{5}{8} P \quad F_{Ey} = \frac{13}{8} P$$

Take beam *DCE*, draw free body force diagram:



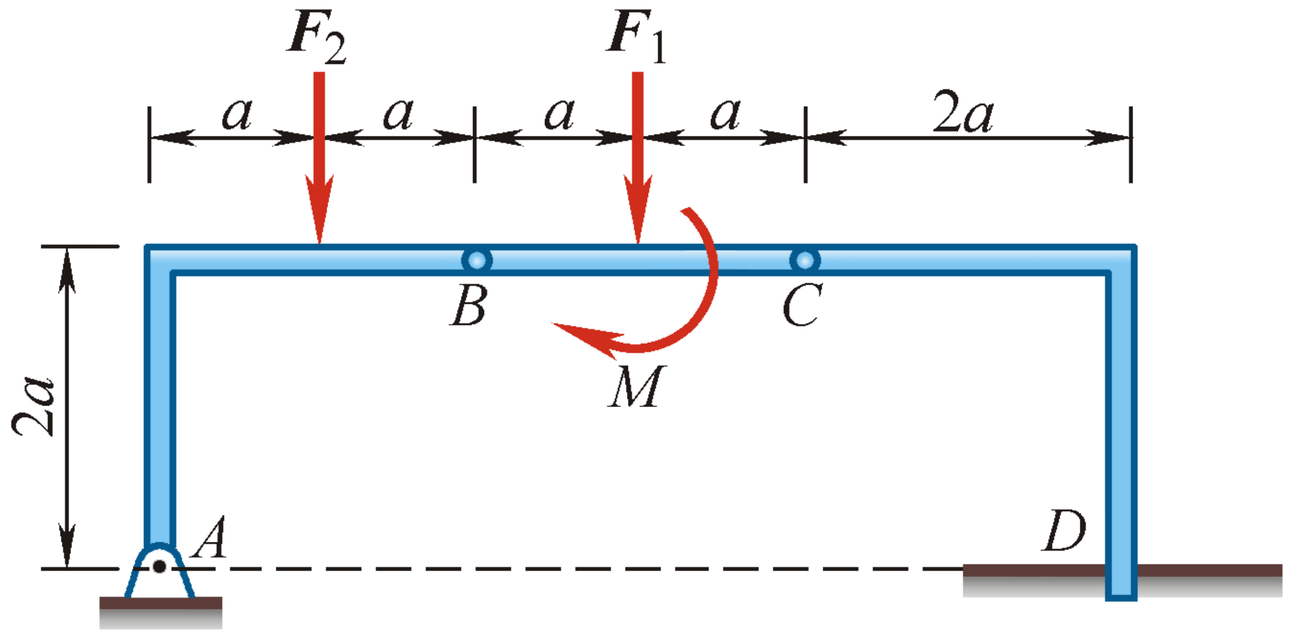
$$\sum M_C = 0 \quad -F_{DB} \cos 45^0 \cdot 2l - F_K \cdot l + F_{Ex} \cdot 2l = 0$$

$$F_{DB} = \frac{3\sqrt{2}}{8} P \qquad \text{(拉)}$$

Example 2-18

Known：**Structure as shown in the figure，** $a, M = Fa, F_1 = F_2 = F$.

Request：**constrain forces at** A ***and*** D. （D **is fixed end**）



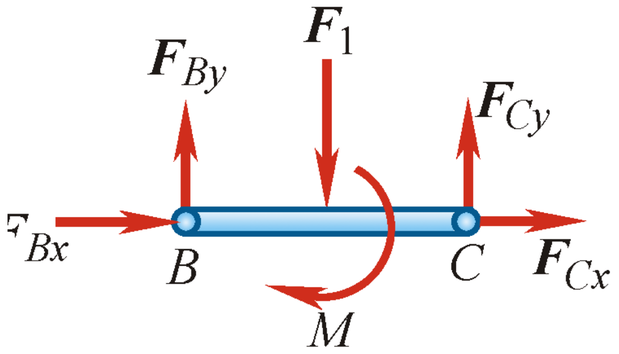
Solution:

Taking ***BC*** as the research object, draw free body force diagram:

$$\sum M_B = 0 \qquad F_{Cy} \cdot 2a - F_1 a - M = 0$$

$$\sum F_y = 0 \qquad F_{By} + F_{Cy} - F_1 = 0$$

$$\Rightarrow \qquad F_{Cy} = F \qquad F_{By} = 0$$



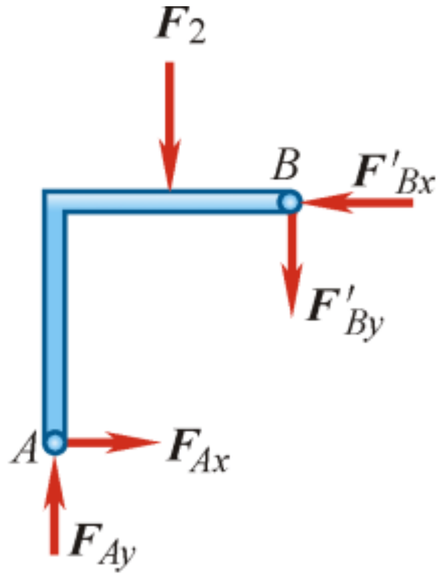
Taking ***AB*** as the research object, draw free body force diagram:

$$\sum M_A = 0 \quad F_{Bx}' \cdot 2a - F_{By}' \cdot 2a - F_2 a = 0$$

$$\sum F_x = 0 \qquad F_{Ax} - F_{Bx}' = 0$$

$$\sum F_y = 0 \qquad F_{Ay} - F_{By}' - F_2 = 0$$

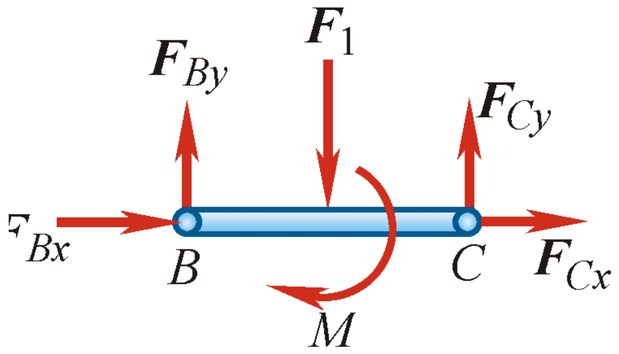
$$\Rightarrow \qquad F_{Bx}' = F_{Ax} = \frac{1}{2}F \qquad F_{Ay} = F$$



Study *BC*:

$$\sum F_x = 0 \qquad F_{Cx} + F_{Bx} = 0$$

$$\Rightarrow \qquad F_{Cx} = -\frac{1}{2}F$$



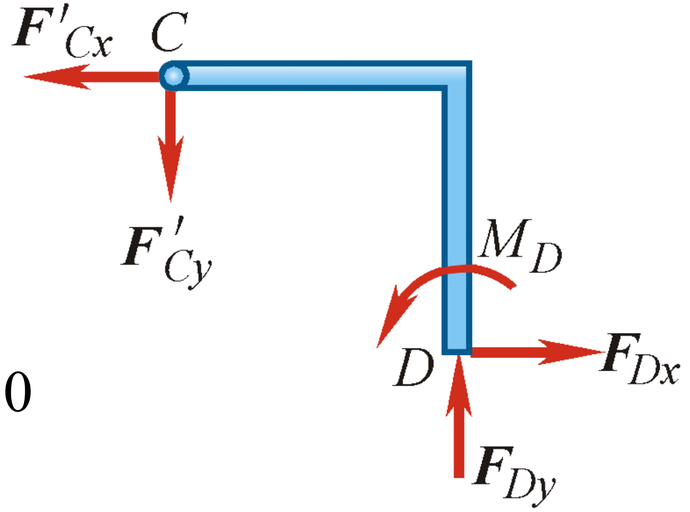
Study *CD*, draw free body force diagram:

$$\sum F_x = 0 \qquad F_{Dx} - F_{Cx}' = 0$$

$$\sum F_y = 0 \qquad F_{Dy} - F_{Cy}' = 0$$

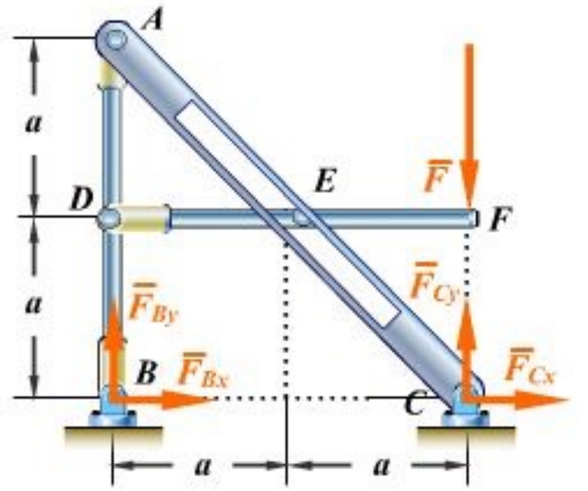
$$\sum M_D = 0 \quad M_D + F_{Cy}' \cdot 2a + F_{Cx}' \cdot 2a = 0$$

$$\Rightarrow \qquad F_{Dx} = -\frac{1}{2}F \qquad F_{Dy} = F \qquad M_D = -Fa$$



Example 2-19 Known：P, a, **disregarding the self weight of each component;**

Request：**constrain force at hinge B.**

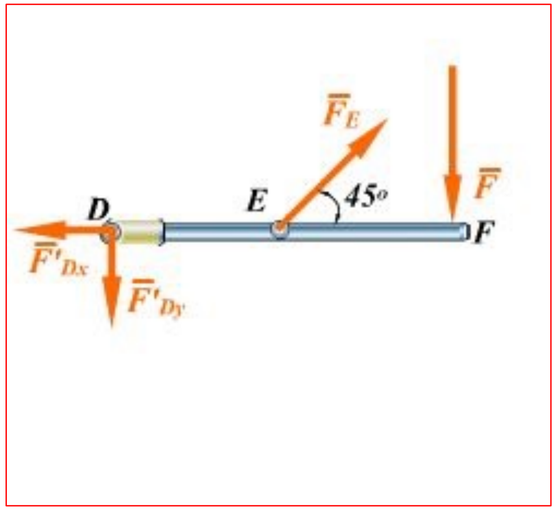


Solution：**Take the entire system, draw free-body force diagram**

$$\sum M_C = 0 \qquad -F_{By} \cdot 2a = 0$$

$$\Rightarrow \qquad F_{By} = 0$$

Study ***DEF***, draw free body force diagram:



$$\sum M_D = 0 \qquad F_E \sin 45° a - F \cdot 2a = 0$$

$$\sum F_x = 0 \qquad F_E \cos 45° - F'_{Dx} = 0$$

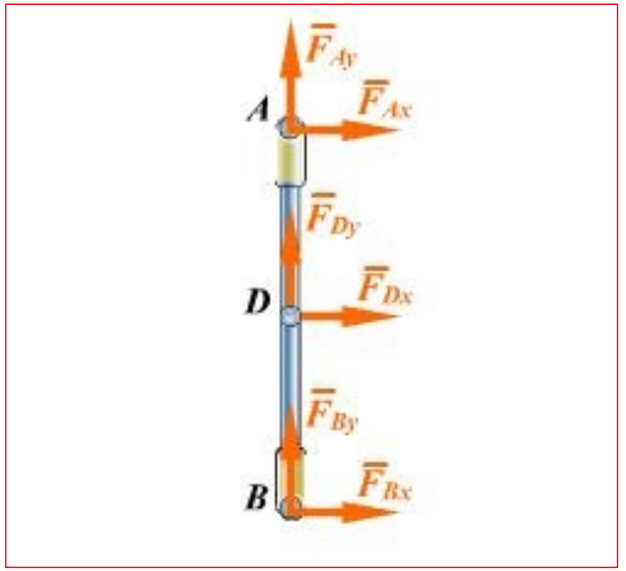
$$\sum M_E = 0 \qquad F_{Dy}' \cdot a - F \cdot 2a = 0$$

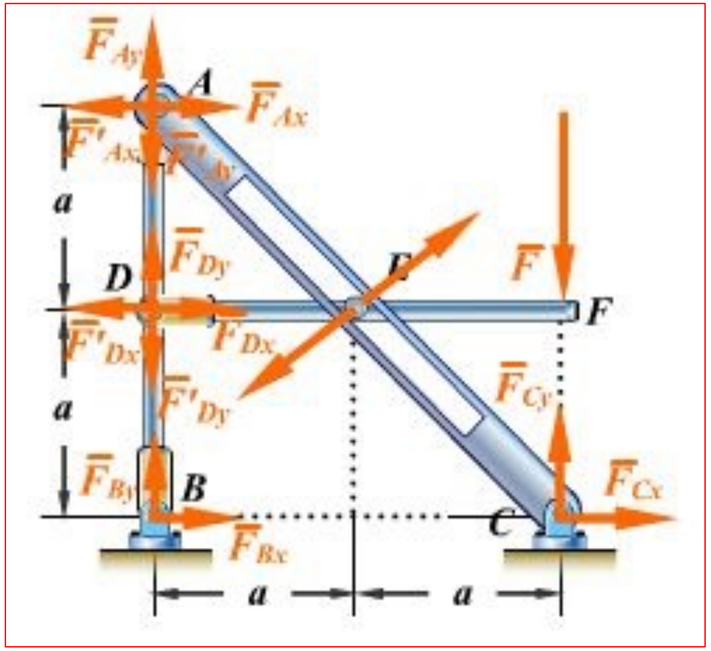
$$\Rightarrow \quad F_E \sin 45° = 2F \qquad F'_{Dx} = F_E \cos 45° = 2F \qquad F'_{Dx} = 2F$$

Study ADB draw free body force diagram

$$\sum M_A = 0 \qquad F_{Bx} \cdot 2a + F_{Dx} \cdot a = 0$$

$$\Rightarrow \qquad F_{Bx} = -F$$

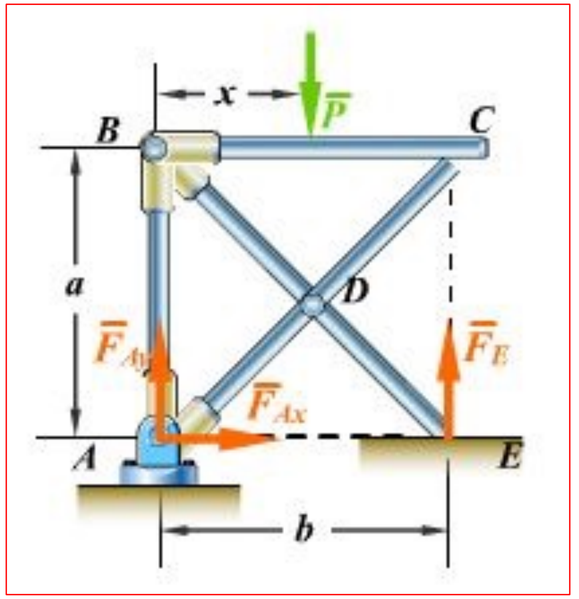




Wrong way

Example **2-20** **Given：** ***a*** **,** ***b*** **,** ***P*****,** ***C and E*** **are frictionless；All members are weightless.**

Prove： The component ***AB*** is always under compression, and the magnitude of the force is ***P***.



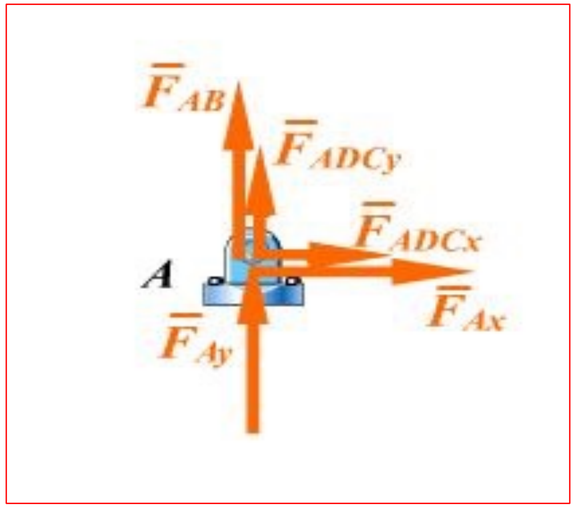
Solve： consider the whole structure， draw the free-body diagram

$$\sum F_x = 0 \qquad F_{Ax} = 0$$

$$\sum M_E = 0 \qquad P \cdot (b - x) - F_{Ay} \cdot b = 0$$

$$\Rightarrow \quad F_{Ay} = \frac{P}{b}(b - x)$$

Take pin *A* as the isolated body and draw the free-body diagram.



$$\sum F_x = 0 \qquad F_{Ax} + F_{ADCx} = 0$$

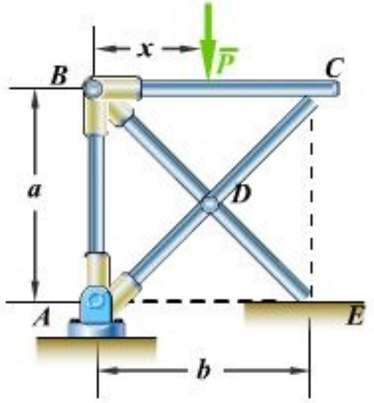
$$\sum F_y = 0 \qquad F_{AB} + F_{Ay} + F_{ADCy} = 0$$

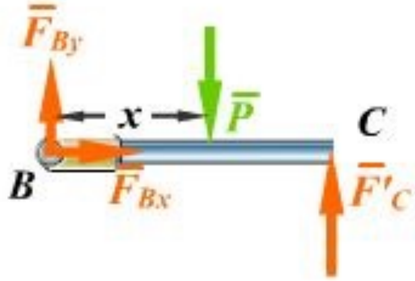
$$\Rightarrow \qquad F_{ADCx} = 0$$

Take *BC* as the isolated body and draw the free-body diagram.

$$\sum M_B = 0 \qquad F_C' \cdot b - Px = 0$$

$$\Rightarrow \qquad F_C' = \frac{x}{b} P$$



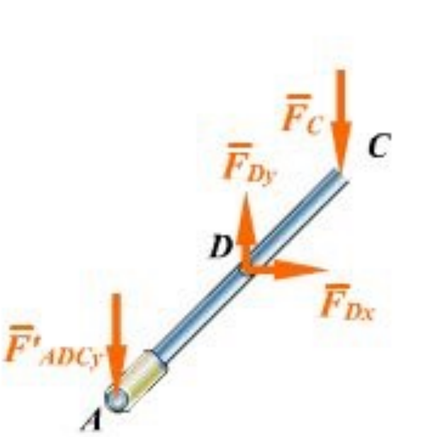


Take *ADC* as the isolated body and draw the free-body diagram.

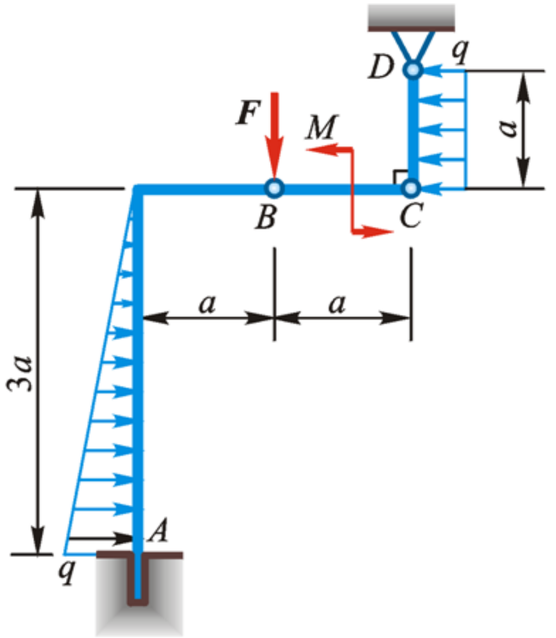
$$\sum M_D = 0 \qquad F_{ADCy}' \cdot \frac{b}{2} - F_C \cdot \frac{b}{2} = 0$$

$$\Rightarrow \qquad F_{ADCy}' = F_C = \frac{x}{b} P$$

$$\Rightarrow \qquad F_{AB} = -P \quad \textit{(in compression)}$$



Example **2-21** Known： $q, a, M, M = qa^2$,



Force F acts on pin B, F = P;

Request: The constraint force at fixed end A and the force exerted by pin ***B on BC and AB rods***

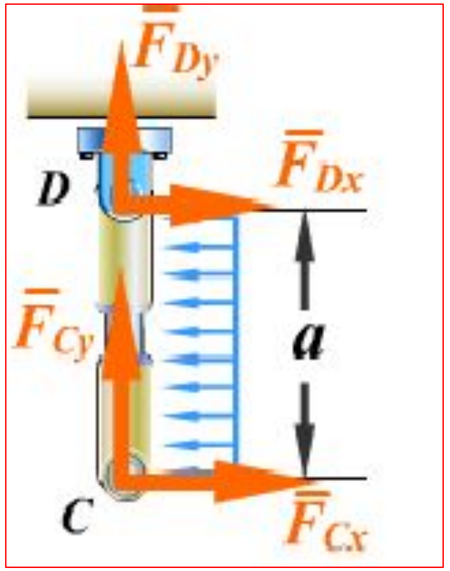
Solution：

Take *CD beam* as research object, and draw free - body diagram:

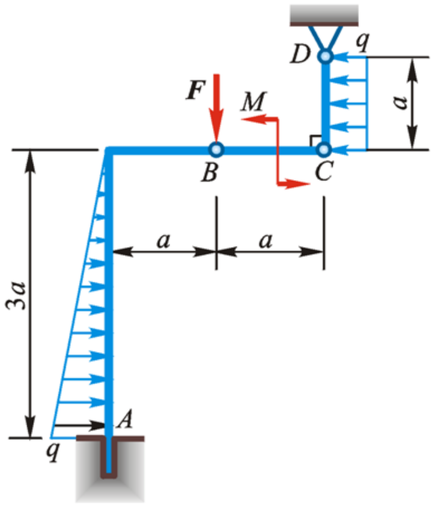
$$\sum M_D = 0$$

$$F_{Cx} \cdot a - qa \cdot \frac{a}{2} = 0$$

$$\Rightarrow \quad F_{Cx} = \frac{1}{2} qa$$



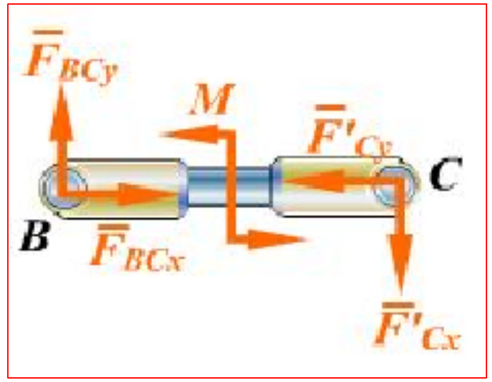
Take beam ***BC*** as the research object（excluding pin B），draw free - body diagram:



$$\sum F_x = 0 \qquad F_{BCx} - F'_{Cx} = 0$$

$$\sum M_C = 0 \qquad M - F_{BCy}a = 0$$

$$\Rightarrow \quad F_{BCx} = \frac{1}{2}qa \qquad F_{BCy} = qa$$



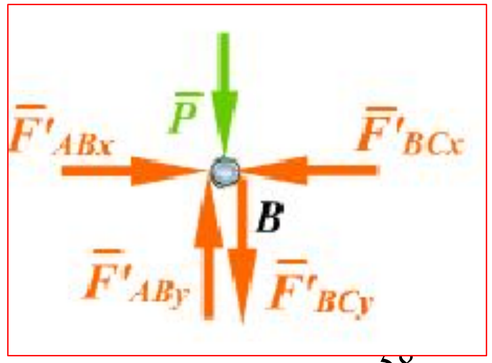
Take pin *B* as research object，draw free - body diagram:

$$\sum F_x = 0 \qquad F'_{ABx} - F'_{BCx} = 0$$

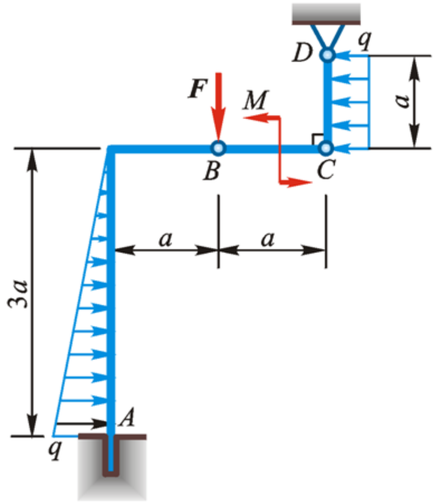
$$\sum F_y = 0 \qquad F'_{ABy} - F'_{BCy} - P = 0$$

$$\Rightarrow \quad F'_{ABx} = \frac{1}{2}qa \qquad F'_{ABy} = P + qa$$

$$\Rightarrow \quad F_{ABx} = -\frac{1}{2}qa \qquad F_{ABy} = -(P + qa)$$



Take beam ***AB*** as the research object（excluding pin B）, draw free - body diagram:



$$\sum F_x = 0 \qquad F_{Ax} + \frac{1}{2}\cdot q \cdot 3a - F_{ABx} = 0$$

$$\Rightarrow \qquad F_{Ax} = -qa$$

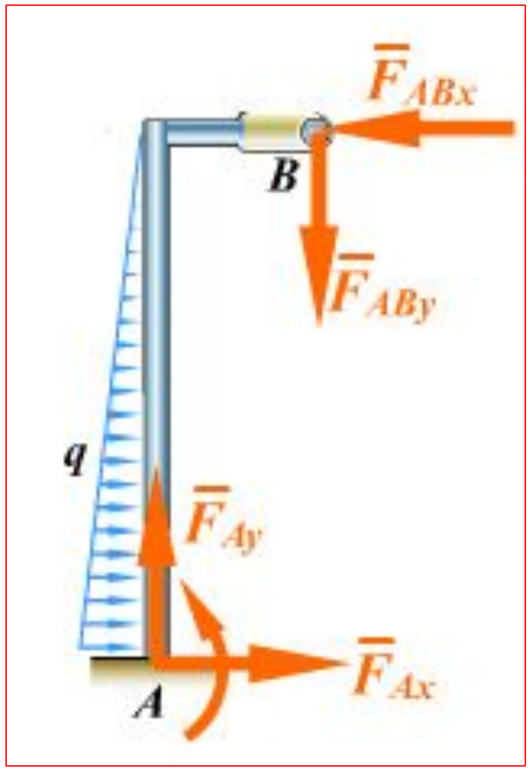
$$\sum F_y = 0 \qquad F_{Ay} - F_{ABy} = 0$$

$$\Rightarrow \qquad F_{Ay} = P + qa$$

$$\sum M_A = 0$$

$$M_A - \frac{1}{2}\cdot q \cdot 3a \cdot a + F_{ABx} \cdot 3a - F_{ABy} \cdot a = 0$$

$$\Rightarrow \qquad M_A = (P + qa)a$$



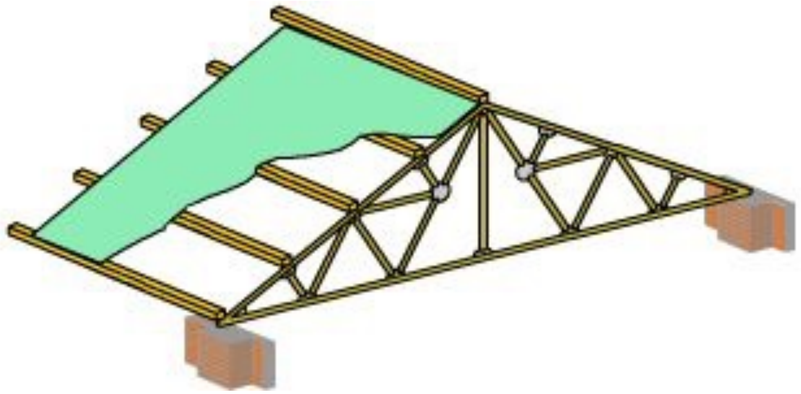
Chapter 2. Planar force system

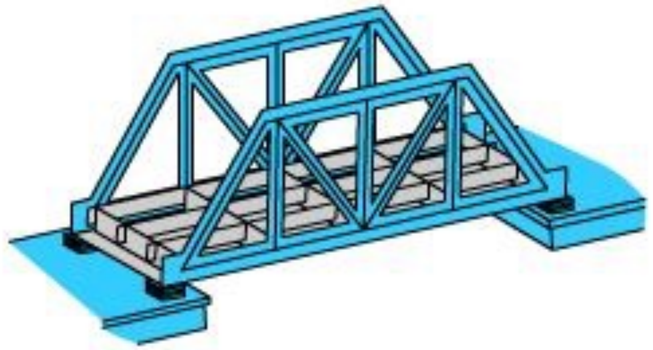
- 2.1 Coplanar concurrent force system
- 2.2 Moment of a planar force about a point
- 2.3 The equilibrium conditions and equilibrium equations for a planar arbitrary force system
- **2.4 Internal force calculation of a simple planar truss**

2.4 Internal force calculation of a simple planar truss

Truss: a structure composed of members connected at their ends by hinges, which maintains its geometric shape under load.

Node: the hinged joint where members of a truss are connected.





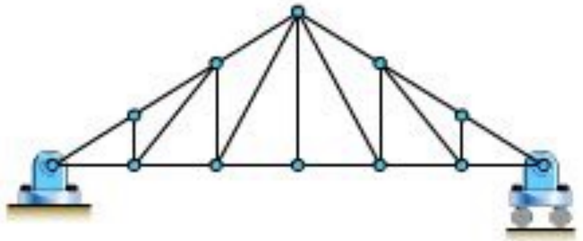
Several assumptions about planar trusses:

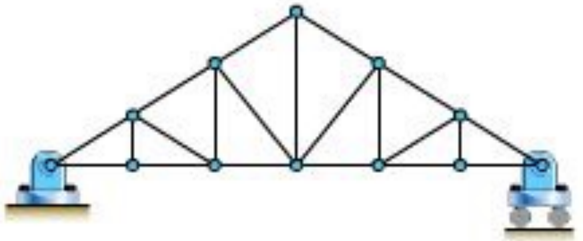
1. Each rod is a straight rod, and the axis of each rod is located in the same plane;
2. Smooth hinges are used to connect the rods to each other;
3. The load acts on the node and is located within the geometric plane of the truss;
4. The self weight of each member is not considered or evenly distributed on the nodes.

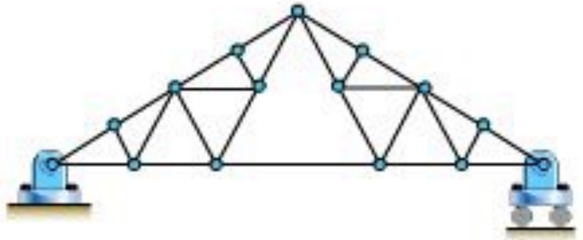


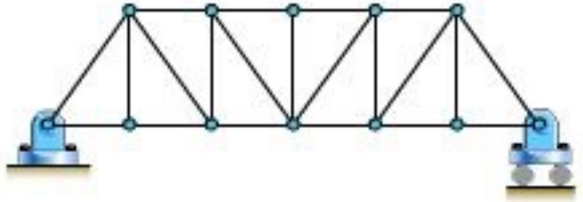
Ideal truss

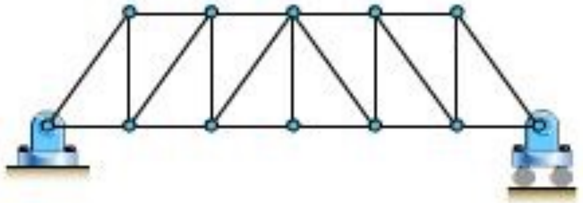
Each member in the truss is a two force rod

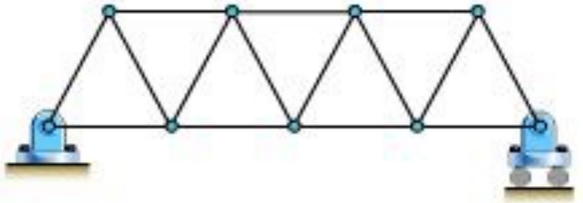


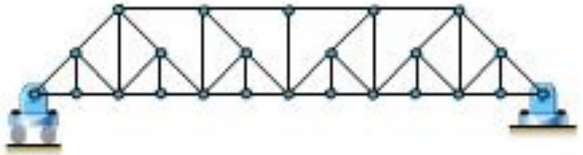


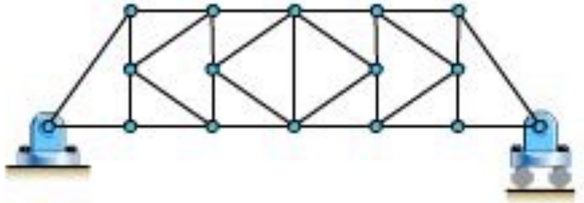




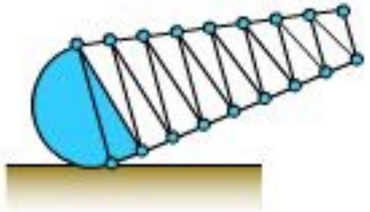


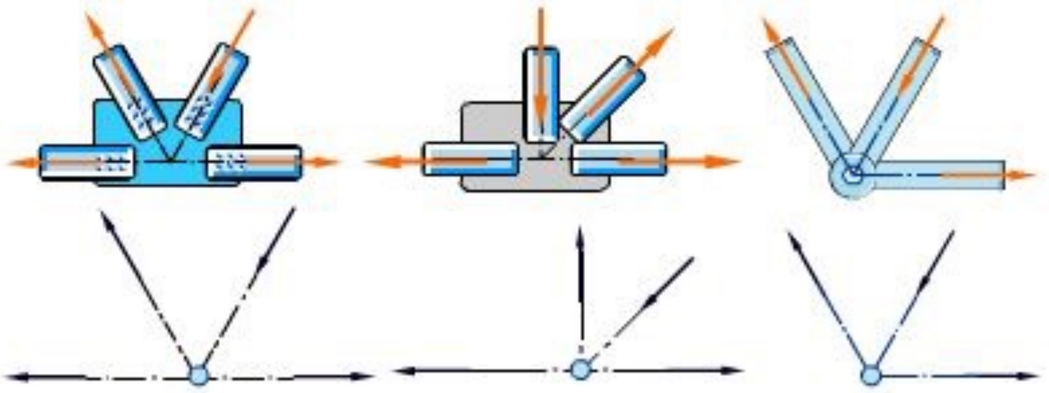








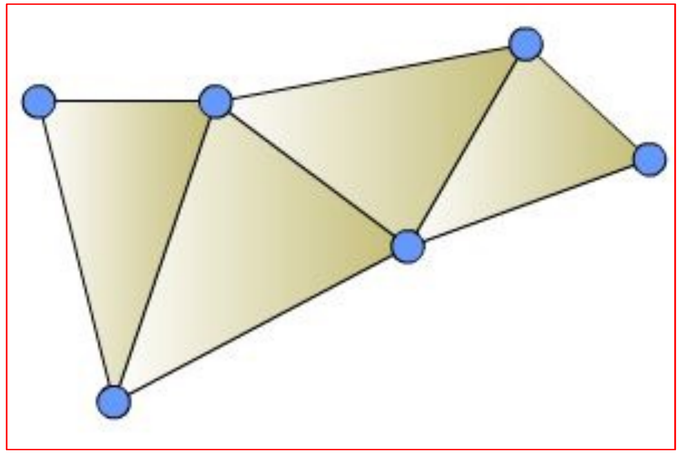


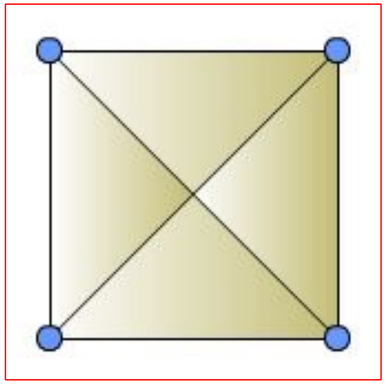


Total number of rods m **Total number of nodes** n

$$m-3=2(n-3)$$

$$m=2n-3$$

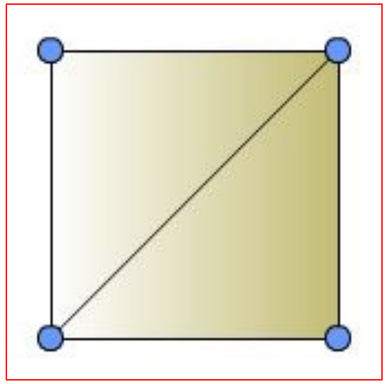




$m>2n-3$

Complex

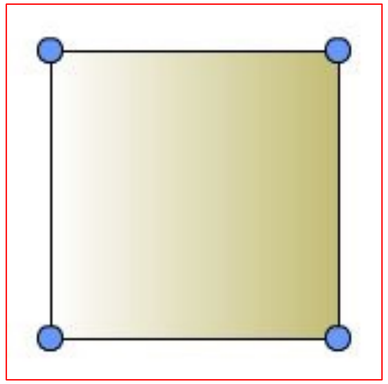
(hyperstatic) planar truss



$m=2n-3$

Simple

(static) planar truss



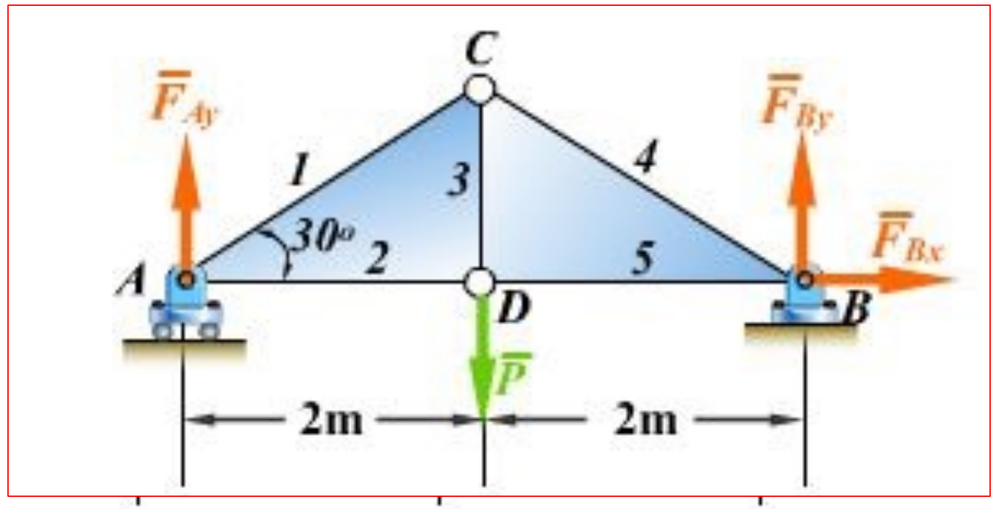
$m<2n-3$

Non-truss

(mechanism)

Method of joints and method of sections

Example 2-22



Known : P=10kN, **dimensions as shown in the diagram. Calculate the force on each member of the truss;**

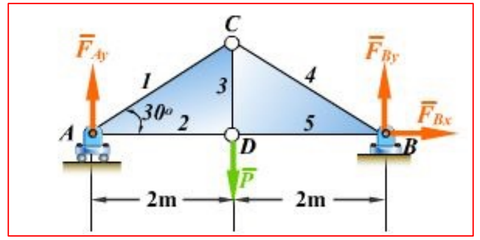
Solution: Take the entire system as research object and draw free body force diagram

$$\sum F_x = 0 \qquad F_{Bx} = 0$$

$$\sum M_B = 0 \qquad 2P - 4F_{Ay} = 0 \qquad F_{Ay} = 5\text{kN}$$

$$\sum F_y = 0 \qquad F_{Ay} + F_{By} - P = 0 \qquad F_{By} = 5\text{kN}$$

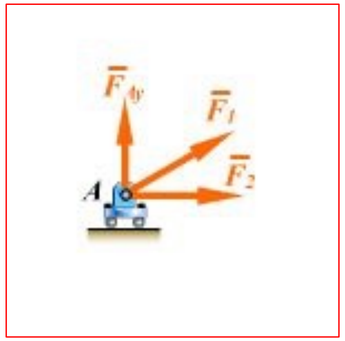
Study node A, draw free body diagram:



$$\sum F_y = 0 \quad F_{Ay} + F_1 \sin 30^0 = 0$$

$$\sum F_x = 0 \quad F_2 + F_1 \cos 30^0 = 0$$

$F_1 = -10\text{kN}$ **(Compression)** $F_2 = 8.66\text{kN}$ **(Tension)**

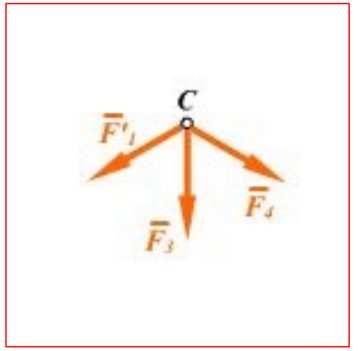


Method of joints

Study node C, draw free body diagram:

$$\sum F_x = 0 \quad F_4 \cos 30^0 - F_1' \cos 30^0 = 0$$

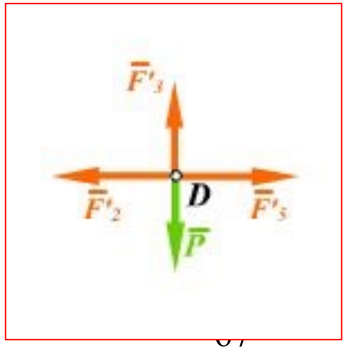
$$\sum F_y = 0 \quad -F_3 - \left(F_1' + F_4\right) \sin 30^0 = 0$$



$F_4 = -10\text{kN}$ **(Compression)** $F_3 = 10\text{kN}$ **(Tension)**

Study node D, draw free body diagram:

$\sum F_x = 0 \quad F_5 - F_2' = 0 \quad F_5 = 8.66\text{kN}$ **(Tension)**

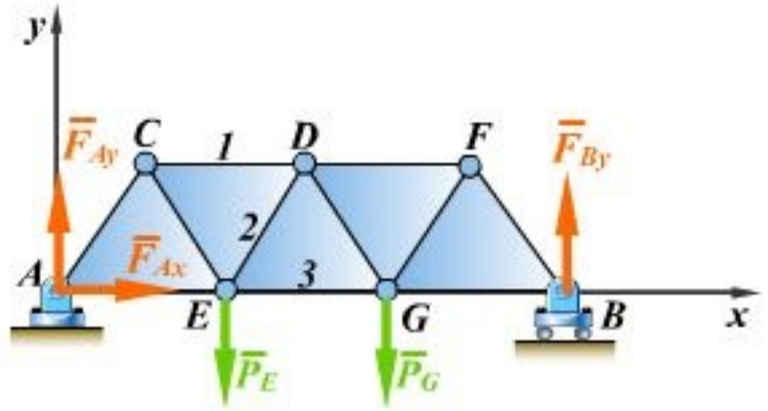


Example **2-23**

Known: $P_E = 10\text{kN},\ P_G = 7\text{kN},$ **Each rod has a length of *1m*;**

Calculate the force on rods 1, 2, and 3

Solution : Take the whole and calculate the constraint force



$$\sum F_x = 0 \qquad F_{Ax} = 0$$

$$\sum M_B = 0 \quad 2P_E + P_G - 3F_{Ay} = 0$$

$$\sum F_y = 0 \quad F_{Ay} + F_{By} - P_E - P_G = 0$$

$$\Rightarrow \quad F_{Ay} = 9\text{kN} \qquad F_{By} = 8\text{kN}$$

Using the method of sections, take the left part of the truss

$$\sum M_E = 0 \qquad -F_1 \cdot 1 \cdot \cos 30^0 - F_{Ay} \cdot 1 = 0$$

$$\sum F_y = 0 \qquad F_{Ay} + F_2 \cdot \sin 60^0 - P_E = 0$$

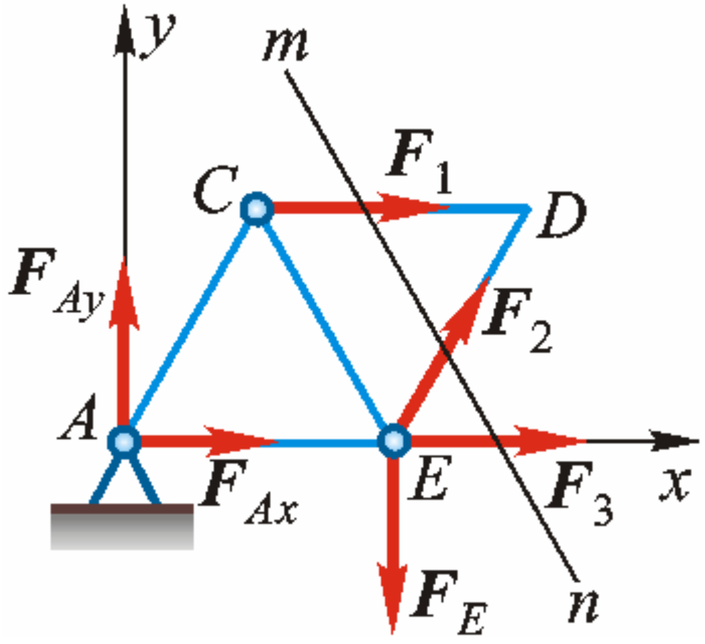
$$\sum F_x = 0 \qquad F_1 + F_3 + F_2 \cos 60^0 = 0$$

Method Of Sections

$F_1 = -10.4\text{kN}$ **(Compression)**

$F_2 = 1.15\text{kN}$ **(Tension)**

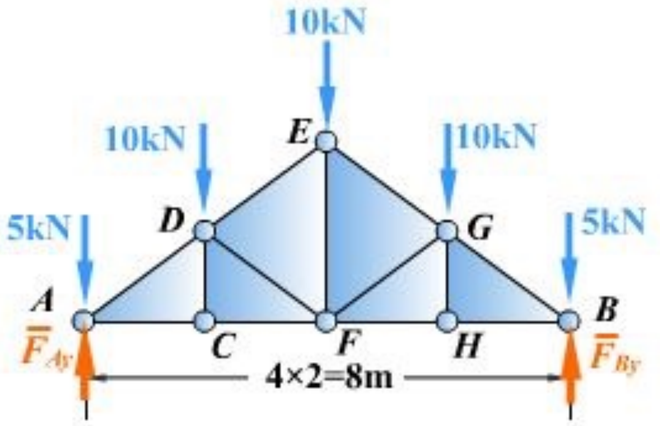
$F_3 = 9.81\text{kN}$ **(Tension)**



Example **2-24**

Known： **Loads and dimensions are shown in the figure**

Calculate: **Calculate the force exerted on each rod**



Solution： Consider the entire system， draw free body diagram

$$\sum F_x = 0 \qquad F_{Ax} = 0$$

$$\sum M_B = 0 \quad -8F_{Ay} + 5\times 8 + 10\times 6 + 10\times 4 + 10\times 2 = 0$$

$$F_{Ay} = 20\mathbf{kN}$$

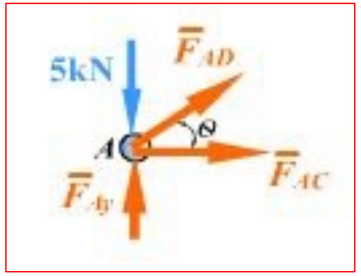
$$\sum F_y = 0 \qquad F_{Ay} + F_{By} - 40 = 0$$

$$F_{By} = 20\mathrm{kN}$$

Calculate the internal forces of each rod

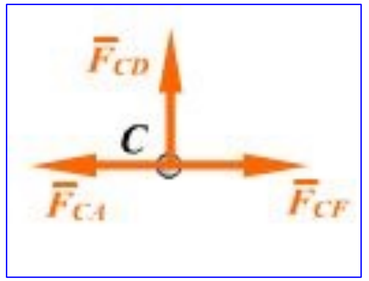
Study joint *A*:

$$\begin{cases} \sum F_y = 0 \rightarrow F_{AD} \\ \sum F_x = 0 \rightarrow F_{AC} \end{cases}$$



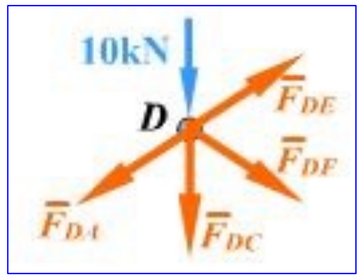
Study joint *C*

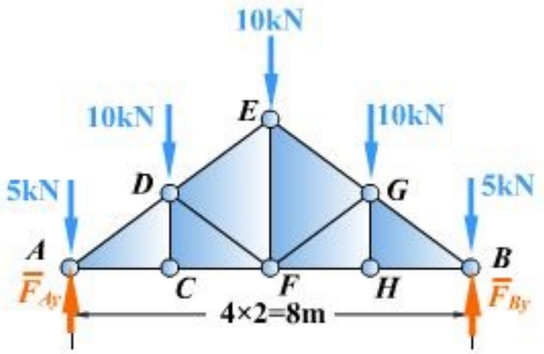
$$\begin{cases} \sum F_x = 0 \rightarrow F_{CF} \\ \sum F_y = 0 \rightarrow F_{CD} = 0 \end{cases}$$



Study joint *D*

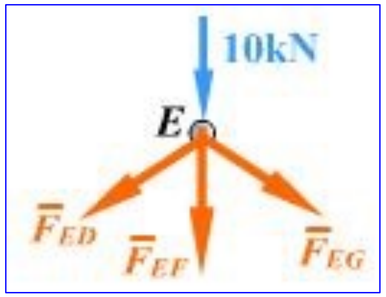
$$\begin{cases} \sum F_y = 0 \\ \sum F_x = 0 \end{cases} \rightarrow F_{DF}, F_{DE}$$





Study joint *E*

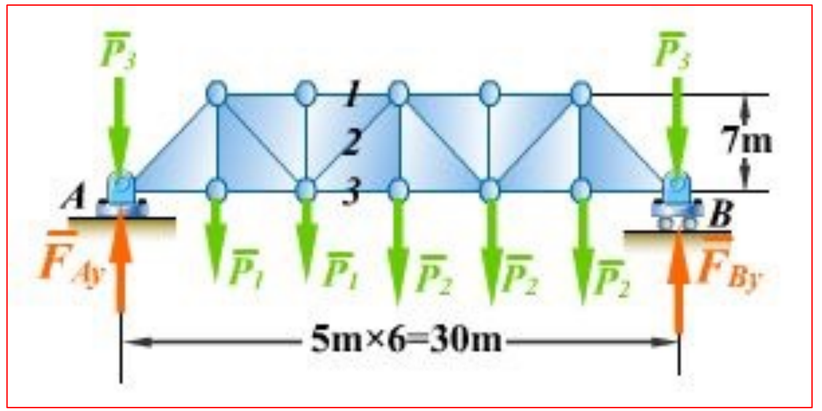
$$\begin{cases} \sum F_y = 0 \rightarrow F_{EG} \\ \sum F_x = 0 \rightarrow F_{EF} \end{cases}$$



Example **2-25**

Known: P_1, P_2, P_3, **dimension are illustrated**

Calculate： Forces exerted on rods **1 ， 2 ， 3**

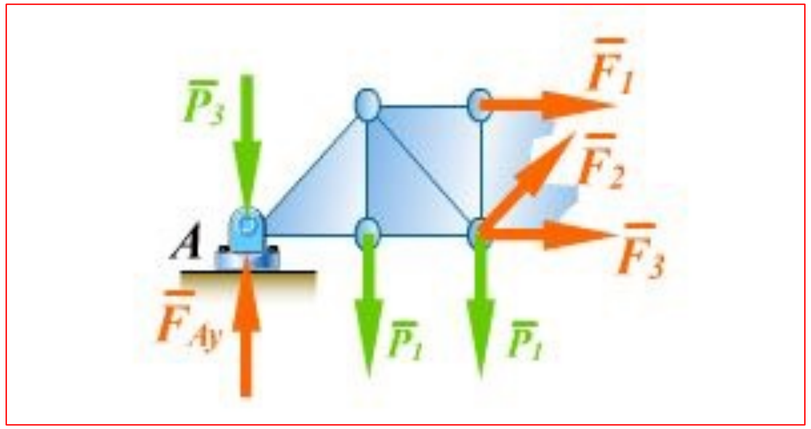


Solution： **Study constraint forces:**

$$\sum M_A = 0 \longrightarrow F_{Ay}$$

$$\sum F_y = 0 \longrightarrow F_{By}$$

Using the method of sections, take the left part of the truss from rods 1, 2, 3

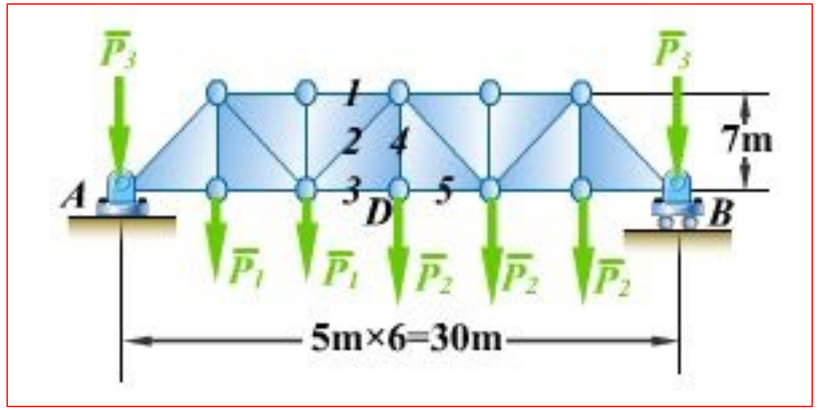


$$\sum F_y = 0 \longrightarrow F_2$$

$$\sum M_C = 0 \longrightarrow F_1$$

$$\sum F_x = 0 \longrightarrow F_3$$

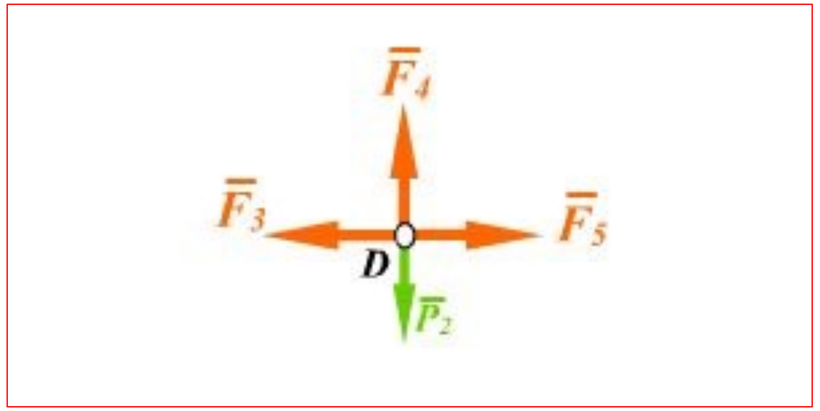
If try to get forces exerted on rods 4,5



Study joint *D*

$$\sum F_x = 0 \longrightarrow F_5$$

$$\sum F_y = 0 \longrightarrow F_4$$



The end of chapter 2